# Aethic Reasoning in Brief:
From a Counterfactual Semantics to the Quantum Measurement Problem

Ajax Benander*

First posted July 2, 2026
This version: September 2026 — preprint

**Abstract**

This paper presents the framework of Aethic reasoning and organizes the dependencies among its semantic, algebraic, and physical constructions. Its central move is that determinacy is a two-place relation between an attribute and an observer's information state, the openness it relativizes being tense-free. What a state leaves open is open rather than unknown, so a uniform, noncontextual, measurement-independent relabeling of openness as ignorance forces a single sample space that the Bell statistics exclude. Whether alternatives exclude one another is content a state carries, which splits superposition into an exclusive and an agreeing regime. A state is admissible only if some proper descendant of it is safe, which is the collapse law and, over a space of paths under a complete and persistent record structure, fixes the quantum–classical boundary as the record partition; collapse is a change of index rather than an event. Pruning ranges over admissible, non-negligible components at every refinement depth. One posited weighting axiom, the path kernel on a fixed finite domain, adds the quantitative face: a Hilbert representation of the admissible clusters, record observables as self-adjoint operators, the squared-norm probability assignment, and the projection postulate as the representation of a stipulated record-compatible restriction. Two representative consequences: the two-tag interferometer exhibits both composition regimes, and an unread which-path record and an unread friend's laboratory fall under one admissibility criterion. The main assumptions are the posited weights and the principle bridging formal invalidity to physical unrealizability; the open physical question is whether a laboratory's records can be kept from escaping, on which the framework's divergence from universal unitarity turns.

## 1 Introduction

Every realist reading of quantum mechanics inherits one debt: saying what unperformed measurements would have shown. Bell and Kochen–Specker close the obvious route to paying it—a standing valuation of the unmeasured—and the dominant Lewis–Stalnaker possible-worlds tradition and standard structural-causal semantics evaluate at complete world or model states, a valuation wearing modal dress. The debt is therefore specific: counterfactual *substance* without counterfactual *definiteness*—truth-apt *woulds* and *mights* with no valuation underneath. That is why a paper about the measurement problem begins with the semantics of counterfactuals, and why the semantics it needs cannot be borrowed.

*The move.* The framework's one move answers the debt: *determinacy is a two-place relation between an attribute and an information state*, never a one-place property of spacetime-located facts, and the openness it relativizes is tense-free. The open state is the *blank*, the framework's central modal posit—structural, occupying the position where classical bookkeeping holds a hidden value, not a gloss on ignorance. And the move's reach is deliberately audacious, because the framework refuses to run two bookkeepings for one primitive: a road collapses into a sinkhole after decades of unobserved erosion, (a *disagreeing* case in the vocabulary of §4, the eroded ground being itself a record—chosen as the move's reach, not its evidence), and relative to every observer on that road the erosion was open—resolving at the moment each learned, never at a moment the ground moved, the unlearned past exactly as open as

*Email: akb9408@rit.edu, Website: https://aethus.org. This paper is the compressed companion to the full treatment, *Aethic Reasoning: A Comprehensive Solution to the Quantum Measurement Problem* [4], where every claim below receives its complete development.

the unformed future, the points unchanged and only the index moving. The case returns in full, guarded against the tensed paraphrase that would tame it, at §2.5.

*Terms.* The framework's vocabulary is small and recurs throughout, so it is fixed here rather than assembled from context. An *Aethus* is an information state: what an observer's records say, with everything they do not say left open. A *blank* is one of those open attributes—open in the constructive sense, not merely unrecorded. Two alternatives stand in *disagreeing superposition* when each states the other's negation, and in *agreeing superposition* when neither does; the difference is the paper's central formal distinction, and it is what ordinary usage of "or" runs together. Agreement is not coherence: agreeing and disagreeing superposition form a qualitative category of Aethic composition, defined before weights or dynamics exist, while coherence and decoherence belong to the quantitative physical realization of that structure, and neither word is a synonym of the other. *Aethic reasoning* is reasoning whose objects are Aethae and whose transformations—refinement, reopening, pruning, weighting—preserve their indexed partiality rather than completing them into total histories or worlds. *Validity* is the property an information state has when some proper descendant of it is safe, none of its own proper descendants forced into contradiction; *pruning* is the removal of the states that lack it, and the *residue* is what pruning leaves. A *child* of an Aethus is a multiplicative extension of it by joint assertion. A *proper child* is a structurally visible component of an admissible declared splitting, a cell of a carving the Aethus holds whose coefficient is nonzero and not measure-zero relative to the rest of the splitting, and a *proper descendant* is a proper child at any depth; that comparison is *relative negligibility*, and at ordinary real weights it means only that the coefficient is positive. Proper childhood is transitive under the algebra companion's admissibility and coefficient-composition assumptions, so it reaches every depth, and the third postulate ranges over it, which is why a valid component carried at negligible weight stays represented but supplies no floor for validity. The algebraic companion owns these relations and states them precisely [10]; later sections refer back here.

*What this overview is.* This is a companion overview rather than an independent research paper. It states the framework's proposal, follows one dependency chain through the program, and gives each principal result with its essential assumptions, its owner paper and its status (Table 1); it claims no result its owner papers do not establish. The chain runs: determinacy is indexed, and blankness is tense-free and constructive (§2.1, §3); composition distinguishes exclusion from openness (§4); the third postulate constrains admissible proper descendants (§§5–5.3); a quantitative axiom supplies the finite physical representation (§9); and the applications show the framework's reach and its limits (§§7–9.1). The counterfactual semantics of §2 is the chain's first worked example, the interferometer of §9.1 its second, and the detector–friend comparison its third.

*The shape of the derivation.* The shape of the derivation, stated in advance so the reader can decide whether to board: blanks are constructive and permanent, so relabeling them wholesale as classical ignorance forces a single sample space—exactly what Bell statistics exclude (§3). Exclusivity between alternatives becomes stated interior content, splitting superposition into disagreeing and agreeing regimes (§4). An information state is valid only if some refinement of it is safe, no record below that refinement forcing a contradiction—the collapse law, its quantifier form derived by elimination and given fixed-point form (§5.2), its physical trigger—that a maintained agreeing superposition and a retrievable record of its alternatives cannot jointly stand—supplied by the union move's identification of interference with record-absence rather than by the elimination; run over a powerset of paths, it draws the quantum–classical boundary as an exact record partition (§7). Collapse is then re-indexing; one flagged weighting axiom supplies the numbers, and with it the three postulates and the union move form an axiom set that represents the Dirac–von Neumann formalism axiom by axiom on the finite path-kernel domain, the observables axiom among its theorems and the projection postulate the representation of a stipulated record-compatible restriction (§9); and one open construction—whether an observer's records can ever be kept from escaping, the irretrievability trigger—is where the framework's divergence from universal unitarity would lie (§8).

*Ownership.* This paper proves nothing its owner papers do not; each result below is attributed to its owner where it is stated, and the ledger of Section 10 lists every claim with its assumptions, its status

Table 1: The principal results, with their assumptions, owner papers and status.

| Result | Essential assumptions | Owner | Status |
|---|---|---|---|
| The counterfactual consequence relation fails Cautious Monotony, so no preferential relation extends it | The dual-peg reading of conjunctive antecedents; finite models with positive weight on every surviving alternative | [6] | Proved; graded representability open |
| Finite declared-descent validity results and the real-weight reduced-form maximum in the specified weighted setting; a partial-filling order under its stated representation assumptions | Compatibility of declared splittings; finitely many attributes; real weights; closure-independent carvings for the partial-filling order | [10] | Proved in those settings; their correspondence with the general reduced form, and an algebra carrying the splitting identities, open |
| Record-partition classification of every cluster | Complete and persistent records; structurally visible record cells; imported record-individuation | [9] | Proved |
| Hilbert representation with Born statistics; projection as Lüders conditioning; no-signaling in the restricted sense of preserved commuting-observable statistics for a fixed record partition | The path-kernel axiom; a stipulated record-compatible restriction | [9] | Proved on a finite domain; ray completeness impossible there for dimension at least two, completeness on an enlarged preparation domain open |
| No noncontextual ignorance reparse of open entries | The four commitments of the determinacy paper; noncontextual, measurement-independent completions | [7] | Proved conditionally; contextual reparses open |
| Detector and friend fall under one admissibility criterion | The record-exclusion lemma; for a laboratory, a physical record premise | [7, 9] | Criterion proved; laboratory premise open |
| The statistical screening structure associated with Reichenbach's principle | Factorized observational states; the exclusive regime | [5] | Proved |
| The decision classification, in two regimes | Supplied roles, binding modes and evaluation laws; the factorization hypothesis | [8] | Proved on its model class |
| The coherent-device exclusion | The would-clause definition; the detector premise | [4] | A proposed divergence, with a test |
| The bridge from formal invalidity to physical unrealizability; the quantum weights | — | [4, 9] | Posited |

and the paper that carries its proof.

Four terms of art[1] will be used and no more—four named objects, not a cap on vocabulary: the primitives among them are the Aethus and the attribute, joined by the primitive union of §4; superposition and the reduced form are defined from those (the reduced form explicitly so, as a residue); the perturbation operator of §2.2—the one operational primitive the semantics adds, and the only thing the paper calls a *modal primitive*—is counted separately on the ledger; and the labels that follow for defined operations and results (reopening, pocket, scope, node, signature partition, the trigger), are shorthands for constructions the text defines in place, each eliminable in favor of its definition— a design commitment against vocabulary inflation, not an accident, with each term introduced only where familiar language provably fails to mark a distinction the argument needs. An *Aethus* (plural *Aethae*) is an observer's complete information state, taken as the index to which reality-attributions attach. An *Aethic attribute* is a question-shaped unit of description—*where is the car parked*, *which slit did the electron traverse*—which is *stated* in an Aethus when that Aethus carries its answer and *blank* when it does not. A *superposition* is what a blank attribute amounts to ontologically, and it comes in two kinds, *disagreeing* and *agreeing*, whose difference part one is largely devoted to (the physicist's anchor: the mixed/coherent distinction rendered as a difference in stated content—the *regime* distinction, to be exact: the attribute side carries a binary flag, declared or undeclared, while relative phase is continuous and enters only through the weighting kernel of §9, so two setups identical on the attribute side can differ in pattern, and no claim is made that the state is attribute-side). And the *reduced form* of an Aethus is the residue of alternatives that survive its consistency constraints. Everything else will be said in plain English, with the full technical development, including the formal definitions this compression rests on, deferred to the long paper.

# 2 Counterfactuals Without Possible Worlds

## 2.1 The Book-Keeping Picture

Begin with an observer at a moment, and collect everything their situation settles: every attribute whose answer their information fixes, directly or by deduction. That totality is their Aethus. It is not a mental state and not a credence function; it is the complete set of facts-for-them, closed under deduction, with every attribute it does not settle left blank. Fixed as objects: an attribute $q$ is a question with an answer space; an Aethus $A$ is a deductively closed assignment of answers to some of the attributes, writing $A(q) = v$ when $q$ is *stated* in $A$ with value $v$, and $A(q) = *$ when $q$ is *blank* (the notation is written for elementary attributes, where unstated and blank coincide; the composite case is separated two paragraphs below). Deductive closure means the books contain their own consequences: if the stated entries of $A$ jointly entail $q = v$, then $A(q) = v$. The block universe is one, and it is rendered relative to this Aethus: one rendering per information state, the observer's history of learning forming a chain of Aethae whose renderings refine one another, never a plurality of spacetimes. A blank attribute's alternatives are carried *on the books* as open entries of that block's description. Where the alternatives themselves reside—whether at other possible worlds, as the classical tradition houses them, or somewhere far stranger—is a question this section deliberately leaves unsettled: for now, everything that follows can be read as pure bookkeeping, and the reader who suspects it is *only* bookkeeping is encouraged to hold that suspicion carefully, because section 3 is addressed to them by name.

Two consequences of the book-keeping picture do early work. First, blankness is *tense-free*: an attribute is blank or stated relative to an Aethus, and where the event it concerns sits in proper time is a separate matter entirely. The observer's unlearned past is exactly as open, on this accounting, as their unformed future—and this shift of blankness's home, from the future alone to anywhere in proper time the index has not reached, is the paper's recurring axis: it powers the sinkhole below, the delayed-choice dissolution of section 7, and the strangeness of the block universe that the collection will price. Where classical metaphysics lets time carry openness, this framework makes openness a fact about the index and lets time carry nothing but geometry. Second, blanks are *constructive*: a blank attribute is not a

[1] Pronunciation, for the subvocalizing reader: *Aethus* follows *aether* ("EE-thus"); the plural *Aethae* takes its "-ae" as in *eye* ("EE-thigh").

gap awaiting a hidden value but a complete informational entry in its own right, encoding the state of the world relative to that Aethus without any determinate value underneath. And blankness is a *retrieval* fact, not a bookkeeping absence: $q$ is blank relative to $A$ when the books leave $q$'s *entire* answer space open, so that $q$ retrieves as all of its states at once. For an elementary attribute—one question, one answer space—that is the same thing as $A(q) = *$. For a composite attribute—a conjunction, which deductive closure makes present whenever its parts are and which is an attribute of the space in its own right—the two come apart, and the reason is its answer space: a composite's answer space is the *product* of its parts', so $X \wedge y$ ranges over four states, not two. With $X$ blank and $y$ stated true, two of the four are open; the composite is unstated, $A(q) = *$, yet not blank but *mixed*, and the postulate of §5 speaks only to the blank grade. (The companion's names: *directly* blank for the mere absence of the composite from the books, *conceptually* blank for the retrieval sense, the mixed status the gap between them. Its dichotomy theorem splits a mixed attribute into stated and blank parts so that the postulate can be applied to each *part*; it confers no status on the whole, and it is the whole that a reopening must return to blank, §2.2.) What would rescue a mixed composite is a *coarsening*: read the conjunction as two-valued, true-or-false, and with $X$ blank it is open over both, hence "blank." That is a presentation of the attribute, not the attribute; the companion's *Decomposition Neutrality*, the principle that a stated-attribute has no privileged decomposition, fixes the attribute's answer space as its parts' product, and the dual-peg reading of the antecedent that this paper adopts is a substantive interpretive choice beside it, not a consequence of it, a rival reading being left open rather than excluded. The lean is booked at §10. The classical habit of hearing "unknown" as "determinate but unrevealed" is precisely the habit this paper exists to break, and section 3 will show where the break is forced — and by what: empirically in the agreeing sector, architecturally everywhere else.

## 2.2 Evaluation by Perturbation

The counterfactual semantics is now three steps, and none of them is travel.

To evaluate *if X had happened, Y would have happened* from an Aethus $A$: first, *reopen* the antecedent attribute—return the page of $A$'s books on which X's attribute is stated to blank, together with everything that rests on it at the chosen scope, and nothing else; second, *restate* the attribute to the antecedent's value; third, *re-prune*—let the framework's consistency constraints (part two's postulates, stated in section 5) eliminate the alternatives that can no longer stand—and *read the residue*. Each step is an operation with a definition. Write $A \ominus q$ for the *reopening* of $q$ in $A$: the maximal deductively closed restriction of $A$ in which $q$ is blank at scope $\sigma$, written $A \ominus_\sigma q$: what "rests on" $q$ is read at the setting, $\sigma$ ranging over three named settings—*evidential-minimal*, *unidirectional* (this paper's default, $\sigma = \text{uni}$, suppressed in the notation where unambiguous), and *bidirectional*—fixed under the second fine point below. Maximality is taken under the declared priorities (laws over particulars, an entry's generality read off its own quantificational form), a policy the owner paper states and whose presentation-invariant derivation remains open. Where those underdetermine—two *particulars* jointly entailing the reopened entry tie, and quantificational form is silent between them—the reopening is *multi-valued*, and the tie enters the residue as co-open pockets feeding the ordinary might/would/grade machinery, as Bizet–Verdi does below. Under-determination is residue structure, not defect. Write $\mathcal{R}(\cdot)$ for the *reduced form*: the set of maximal consistent refinements surviving the pruning, each refinement a pocket of co-open content, the whole carrying a weight measure $w$. The refinements settle each blank to one of its values: the blanks of the semantics are scenario blanks, each attribute's single-peg carving belonging to every evaluation state's standing structure, so that a blank is the union of the attribute's determinate states rather than a bare filling of open values [6]. The *perturbation* by an antecedent $X$ (stating attribute $q_X$ to value $x$) is then

$$\pi^{\sigma}_{X}(A) \;=\; \mathcal{R}\big(\; (A \ominus_\sigma q_X) + \{q_X{=}x\}\;\big),$$

and the modal operators receive their truth conditions directly on it:

$$
\begin{aligned}
A \models X \,\Box\!\!\rightarrow Y \quad &\text{iff} \quad \text{every pocket of } \pi_X(A) \text{ settles } Y \text{ affirmatively,} \\
A \models X \,\Diamond\!\!\rightarrow Y \quad &\text{iff} \quad \text{some pocket of } \pi_X(A) \text{ does,} \\
\mathrm{P}_A(Y \,\|\, X) \;&=\; \frac{w\big(\pi_X(A) \restriction Y\big)}{w\big(\pi_X(A)\big)},
\end{aligned}
$$

where $\pi_X(A) \restriction Y$ collects the $Y$-affirming pockets. *Would*, *might*, and *the probability that* are thereby one operator read at three grains—universal over the residue, existential over it, and its weighted proportion—and where the residue splits, the framework does not force a binary verdict but reports the graded fact.

Three fine points fix the operator's behavior.

*Two operations: indicative selection and subjunctive reopening.* First, the perturbation is two operations selected by mood, not one rule read at two limits: under the *indicative* setting the evaluation is selection over the standing residue, the pockets in which the antecedent holds and nothing else; under the *subjunctive* settings it is computed from restrictions of the state itself, the priority-selected maximal closed blank restrictions, one or several as the priorities select or tie, each restated and re-pruned with the residue the union over them, and never by completing the state and reopening its pockets one by one, which computes a different operation (the companion's three-attribute example shows the two can differ in verdict). The two moods share the pruning and the weighting and are otherwise distinct operations; no coincidence condition between them is claimed, and the tradition's "two modes" are these two operations, not limits of one rule. For the *agnostic* Adams evaluator—death stated, shooter blank at weight $w_O$ on Oswald—the subjunctive reopens the shooter in the state itself at posterity scope, the recorded posterity of the shot clearing with it, the death included, in every pocket at once; with no conspiracy on the books nothing regenerates the death, so the would comes out *false*, the might holds through second-shooter pockets carried at small positive weight $w_S$, visible against the rest, and the grade is small and positive, $w_S$ against the whole residue, for every $w_O$; pockets at exactly zero weight would be no proper children and would sustain neither reading. The interpolating grade $1 - w_O$ that a pocket-wise procedure produced belonged to that procedure and is withdrawn with it; what an evaluator all but certain it wasn't Oswald means is carried by the indicative, which returns true at grade one. Two commitments of the operation follow and are owned: centering fails for stated antecedents, since exempting a stated antecedent from reopening would privilege the actual outcome over its alternative, the toss that landed heads over the landing tails, and no postulate supplies that privilege, the alternatives being symmetric; and credence enters through the indicative alone, the subjunctive's grade tracking only the standing weight of what the reopened state leaves. A mood-sensitive graded verdict for uncertain evaluators is not something a binary semantics produces. (The previously run checks—streetlight, Morgenbesser, Bizet–Verdi, backtracking—all have uniform pockets, so their verdicts are unchanged; and under the indicative setting the operator computes conditional probability *outright*, since selection followed by renormalization is conditioning by the theorem of section 6. Adams's thesis—that indicative credence goes by conditional credence—thereby arrives as an *identity of the indicative setting* rather than a constraint imposed on it [1] — placement, per the column discipline of section 6.) Lewis's "small miracle"—the minimal law-violation that manufactures a foreign world—becomes an auditable bookkeeping operation: no world is manufactured; a page of the books is reopened at a named place.

*The scope convention.* Second, the reopening's *scope* is the semantics' one declared conventional parameter, and it is structured: one parameter, three named settings, selected by mood. *Evidential-minimal* clears the antecedent's entry and its deductive cone only, keeping whatever enjoys independent support. (The indicative is none of these settings: it selects within the evaluator's own Aethus rather than reopening it.) *Unidirectional* clears the antecedent together with the generics wired to it on the direction convention's forward side (its recorded effects; the direction an emergent, perspective-relative reading mined from the attributes' disjointness content [4])—the standard subjunctive's setting and the default. *Bidirectional* runs the clearing across every wired generic, direction-blind, both sides at once—the backtracking setting (section 2.4). Mood selects the operation, and for the subjunctive the

setting; the reopening acts on the state itself at that scope, never pocket by pocket. Two operations; scope is the subjunctive's one declared conventional parameter, stated, visible, and movable rather than hidden inside a metric, and booked on the closing ledger as a convention, since it does verdict-determining work; one further clause was booked beside them in earlier prints and is now recorded as a reading rather than a convention: conjunctive antecedents reopen *conjunct-wise*—each conjunct's attribute returned to blank, its ground cleared—which, under the dual-peg reading of conjunction, a substantive choice of antecedent interpretation, is a consequence of the three steps as defined, once blankness is read at the grade section 2.1 fixes and the antecedent is read by the *dual-peg reading*, each conjunct's peg a separate argument of the reopening. That reading is the framework's reading of "and" in an antecedent; the two-valued attribute *is* $X \wedge y$ *true?* is a coarsening of the composite, a different attribute, and on it nothing is cleared. The first step reopens *the antecedent attribute*—returns it to blank, in the retrieval sense: its whole answer space open, its carving kept, since the carving states no value and the maximal restriction retains it, so that the reopened attribute is a scenario blank whose pockets are its values—and only then does the second step restate; the composite conjunction is that attribute, since the operator takes the antecedent as its argument and closure makes the conjunction an attribute of the space, and its status is read of it rather than delegated to a part (section 2.1's one lean on the companion's Decomposition Neutrality, which the companion derives at length [4]). So, on the dual-peg reading, a conjunctive target is returned to blank only with every conjunct blank; a rival reading of the antecedent is not excluded by Decomposition Neutrality, which fixes the attribute's decomposition and not the antecedent's interpretation. Reading the conjunction as one attribute and blanking it *maximally*—keeping an already-actual conjunct and its ground because blanking the other already unstates the conjunction—leaves the target directly blank but *mixed*, one part present, which is not superposition at all; it is rescued only by calling the target open insofar as its blank component is, and that delegation of the whole's status to one part is what section 2.1's reading of a composite excludes; the Cautious Monotony result of section 2.3 runs on the derived reading—stated here, graded there.[2] Third, the inventory: the semantics adds exactly *one* new modal primitive, the perturbation operator itself; validity, weight, and the pruning are all inherited from machinery the framework needs for independent reasons. Rounding that one down to zero would be a refutable boast, and the theory is stronger for admitting it.

*The dependency structure, booked.* And one further piece of equipment must be booked while we are counting: the reopening is well-defined only because the books carry a *dependency structure*—which entries rest on which—so that $A \ominus q$ has a determinate answer (set-valued where the declared priorities tie or rank incomparable maxima). For a bare deductively closed theory it would have none at all: $\{p,\ p \to q\}$ can lose $q$ by dropping either member, the remainder-set problem the belief-revision literature was built around [2]. That dependency structure is extra structure, it is exactly where an ordering-theorist will look for the deleted ordering's ghost, and it therefore goes on the ledger.

Three features distinguish this from the possible-worlds treatment at the level of machinery, before any philosophy is spent. The evaluation never leaves the evaluator's own structure: the perturbation is an operation on the books, not a search across worlds.[3] The evaluation is *checkable*: each alleged

[2]The Adams pair falls out of the scope menu, and each sentence is best run from the evaluator who would naturally assert it—a small display of the indexing doing pragmatic work. The *agnostic's* indicative—*if Oswald didn't shoot him, someone else did*—runs evidential-minimal by selection: the death, independently supported, stays stated, the surviving pockets are the someone-else ones, and the indicative comes out true (the *believer's* indicative, having no pocket in which the antecedent holds, has an empty selection and no indicative verdict, the counterfactual there being the subjunctive's). The *believer's* subjunctive—*if Oswald hadn't shot him, someone else would have*—runs unidirectional by reopening: the death, downstream of the shot, is cleared, and absent a stated conspiracy the would fails while the mights survive. Morgenbesser's coin still passes for free on the subjunctive setting (the coin is not in the bet's causal posterity, so it stays fixed). Two operations throughout, mood selecting between them.

[3]Every documented language runs modality through one shared determinate ledger, as though each speaker were attached to a single complete fact-state with possibility bolted on as an operator over it; Aethic reasoning reverses the attachment at the person—you are attached to your *partial* information state, blanks and all, and the first-person pronoun points at the Aethus, not through it. The cost of that reversal is priced as the third cost of §10; the companion assembles the typological and developmental evidence [4].

dependence of Y on X is tested by a single perturbation, with the verdict read off the residue, so causal claims become audit-shaped rather than similarity-shaped. And the evaluation is *indexed*: the same counterfactual can receive different verdicts from different Aethae without contradiction, because the verdicts are facts about differently-informed books, not competing claims about one ledger.

### 2.3 The First Headline: The Ordering Is Deleted, Not Refined

The Lewis–Stalnaker tradition [29, 37] evaluates the same sentence by locating the closest possible worlds where X holds and asking whether Y holds there, with *closest* supplied by a similarity ordering over worlds. The comparison is sharpest at the level of the clauses themselves. Lewis's official truth condition, in its standard form:

$$w \models X \mathrel{\square\!\!\rightarrow} Y \quad \text{iff} \quad \text{some } (X \wedge Y)\text{-world is closer to } w, \text{ by } \leq_w, \text{ than any } (X \wedge \neg Y)\text{-world,}$$

(vacuous truth aside), where $\leq_w$ is the similarity ordering centered on $w$. Set it against the clause of section 2.2 and perform the visual audit: the evaluation point changed type, from a world $w$ to an Aethus $A$; the search across worlds became an operation, $\pi_X$, on the evaluation point's own structure; and *the parameter $\leq_w$ appears nowhere on the right-hand side.* Fifty years of refinement adjusted that parameter's weights—laws over particular facts, perfect match over approximate—without dislodging the structural complaint that the ordering is imposed rather than derived: nothing in the picture says why similarity, or which similarity, tracks the truth of counterfactuals. A reader from the belief-revision literature will recognize mined ground here, and it is walked knowingly: Lewis himself proved that premise semantics—evaluation by consistent adjustments to a premise set, no ordering in sight—is equivalent to ordering semantics [30], Grove proved the analogous representation theorem for belief revision [25], and the perturbation is, formally, contraction-then-expansion in the AGM mold [2]. Non-appearance of the parameter is therefore not the standard; non-recoverability is. The paper's considered claim is calibrated accordingly: the ordering's work is *derived from and localized in* the deductive dependency structure of the books, rather than free-floating over worlds—and the recoverability contest now stands split, one stage closed and one posed. Closed: the would-clause's consequence relation *violates Cautious Monotony*, by the countermodel displayed at the end of this paragraph (its natural-language face an ordinary explaining-away pattern, and the companion separates the cases cleanly [4]: the bare conjunction *if it had rained and the lawn had been wet* should leave hearers divided (a prediction, not yet a report: no elicited-judgment study of this case exists)—many keeping the sprinkler run that actually wetted it, so that the gutters still overflow—while the *anyway*-variant shows the reopened parse is grammaticalized in ordinary usage, a marker whose whole job is to reopen a conjunct's ground; the countermodel runs on the *uniform reading*, under which a conjunctive antecedent is reopened whole, each conjunct's ground cleared even where a per-conjunct application of the pocket-wise selection rule would pass an already-holding conjunct through — given the dual-peg interpretation of conjunctive antecedents, reopening must blank each constituent attribute, which section 2.2 derives from that interpretation rather than stipulates — so the headline is graded: *closed, on the dual-peg reading of conjunctive antecedents*—a substantive interpretive choice that Decomposition Neutrality does not itself supply, as the companion now states (section 2.2, with the derivation's proofs in the companion [4])—with whether the bare natural-language conjunction defaults to that reading a question about the language—exactly what the instrument already in the literature [12] adapts directly to adjudicate—and not about the operator); since preferential relations satisfy Cautious Monotony, and ranked relations—those a similarity ordering represents—are preferential, the relation agrees with no smooth preferential relation on all of its evaluable verdicts, so it is not preferential and *a fortiori* not ordering-representable at the consequence-relation level—the Lewis and Grove equivalences notwithstanding, proved as they were for single-output revision on unindexed, binary-verdict theories (the Kraus–Lehmann–Magidor audit, the countermodel, and the surviving core—Reflexivity, Right Weakening, And, Cut wherever the reopenings are nested, Left Logical Equivalence at attribute grade, Or—are the companion's [4]).

*The countermodel, in outline.* Four attributes suffice—rain, the sprinkler, the lawn's wetness with the sprinkler as its ground, and overflowing gutters—and the failure sits at one place: a conjunctive antecedent reopens a conjunct's ground and so resurrects pockets the plain antecedent pruned. The

clearing is not an exception to downstream-only clearing. Reopening both conjuncts *and* clearing their supports is derived, on the dual-peg reading, as the only reading that returns the composite to blank, since under closure a support left stated would simply re-derive its conjunct; what remains empirical is only whether the English conjunction tracks this operator. The books, the laws, the derivation, and the residues are the companions' [4, 6].

The witness is stated on the dual-peg reading of a conjunctive antecedent, on which a conjunctive target is returned to blank only with every conjunct blank; that reading is a substantive commitment of the semantics, and a rival reading of the antecedent, blanking the conjunction as one attribute and keeping an already-actual conjunct, is not excluded by Decomposition Neutrality, which fixes the attribute's decomposition and not the antecedent's interpretation. On the rival reading the witness does not arise. The non-preferentiality result is therefore scoped to the uniform dual-peg reading, and the semantics paper carries the comparison of readings and the graded-representation question it leaves open [6].

Two further exhibits complete the deletion's formal face, at one sentence each. *Completion-blindness*: two states identical in every stated attribute and every weight apart from the cross attribute over a blank's alternatives, declared false in one and left undeclared in the other, possess identical completion classes — the completion map is a projection that discards exactly that information, every completion resolving the blank and destroying the co-presence the entry describes — while splitting the verdict of a conditional keyed to that attribute; no function of a state's total completions computes the verdicts, supervaluation included, so the classical vocabulary is not re-parameterized but expressively short. And *imaging*: the tradition's own probabilistic repair — Lewis's 1976 proposal, moving each world's probability mass to its closest antecedent-world — consumes the similarity ordering pointwise, the transport plan being the ordering itself, while the graded verdict here routes the mass through the blank intermediate, consulting dependency and validity only; the deletion extends to the probabilistic tradition's own fix.

The Aethic replacement should not be described as a better ordering. No world-similarity ordering is primitive. Its work is split among structures that were independently needed: the *dependency structure* of the books, which fixes what a reopening holds; *validity*—the binary fact of whether an alternative survives the re-pruning at all—and *weight*—the graded structure on what survives. Nothing ranks the surviving alternatives against an ideal; they survive or they do not, and among the survivors weight measures rather than ranks. This is the paper's first headline because it is a deletion with an address: the parameter philosophers spent five decades tuning is identified as an artifact of housing alternatives elsewhere, and it vanishes the moment the alternatives come home.

The deletion lands twice: in the formal semantics of natural-language modality, where Kratzer's ordering source is the parameter removed, and in the belief-revision tradition, where the Ramsey test's addition against revision is the indicative–subjunctive seam seen from its own side [28, 22]; both are developed in the semantics paper.

### 2.4 The Standard Battery

The adversarial cases the tradition built fall to the two operations without additional machinery, and the semantics paper works each [6]: Bizet–Verdi comes out as a might-tie with the would failing, since the reopened residue holds both nationalities; Morgenbesser's coin passes on the subjunctive setting because the coin lies outside the bet's causal posterity and stays fixed; the Adams pair divides between the operations, the indicative selecting and the subjunctive reopening, which is where credence enters through the indicative alone; backtracking is the scope parameter's posterity setting; and the continuous case, which a whole-world similarity ordering cannot grade, is graded by the standing weights of the pockets the perturbation leaves. None of these is an axiom of the semantics; each is the operation run on the case.

### 2.5 Worked Checks, Compressed

Two examples carry the semantics' weight in the long paper, and they compress well.

*The streetlight.* An observer drives through an intersection; the light was green; had it been red,

the driver would have stopped. On the unidirectional setting the reopening clears "proceeded" as recorded posterity of the light, restates red, and re-prunes to the alternatives in which the standing disposition meets red and the car stops: would-stopped at grade one, the flanking facts untouched. On the evidential-minimal setting the proceeding entry survives the narrower clearing and closure then excludes red, so no admissible intermediate exists and the run is infeasible rather than vacuous, which is the scope parameter doing its work.

*The sinkhole.* A driver is stopped at a red light; two minutes later a sinkhole opens just ahead, where the car would have been on a green, the erosion behind it having run underground for decades, unobserved. Relative to the driver's later state, which states the sinkhole and its history, *had the light been green, the car would have met the hole* holds; relative to the state at the light, where the sinkhole and its history are blank, it does not, since that state's continuations include ones in which the car crosses the same ground with no hole beneath it. The collapse produced a record, but a record's existence resolves nothing for an observer until that observer's successor state comes to state it; and the later verdict does not replace the earlier one, which remains true relative to the earlier state, the erosion's temporal precedence supplying no determination at that index. The unlearned past is as open as the unformed future. That is tense-free indexing in one case. The same indexing blocks the recursive construction of one determinate history: dependencies established at different indices compose only once a common admissible index is established, both survive transport to it, and a composition rule is given, so what is blocked is the unlicensed construction and not a common mechanism history where those conditions hold. Both are run in full in the causal companion [5].

# 3 A Note Against Deflation

Before anything further is built, one objection must be met, because everything depends on how it is met—and the meeting is a pricing, not a refutation: empirical where records run out, architectural everywhere else. The objection is aimed at everything granted so far, sections 2.1 through 2.3 taken together, and the note is accordingly a *conditional* result—granted that ledger, deflation fails—which is the form that makes it auditable. The objection says: all of the above is classical metaphysics under new labels. Call attributes "blank" if you like; a blank is just an unknown; an unknown has a determinate value behind it; your "superpositions" are ignorance with a vocabulary. If this deflationary reading were available, part one would be a notational variant of the possible-worlds picture it claims to replace, and part two would be sleight of hand.

The reading is available only at a price, and the argument has one fork. The target is exact: a deflation that keeps what a state leaves open as constructive openness and adds a determinate value at the same index contradicts the state's own content, which is the internal preservation constraint; a deflation that instead translates openness into ignorance uniformly, noncontextually, and with measurement independence posits a lossless assignment whose image is a single sample space, and that image is what the Bell statistics exclude, under the commitments D1–D5 stated in the determinacy paper [7]. Neither horn uniquely selects information-state indexing: a contextual assignment, and a deflation confined to the classical sector where the statistics never bite, lie outside the targeted class and are not eliminated by the obstruction. What the framework claims against them is architectural, that one semantics applied uniformly across the classical and quantum sectors is cheaper than two bookkeepings with a cut between them, and that claim is a cost comparison the reader is free to dispute, not a theorem.

The closure earns the headline the introduction opened on: the framework provides *counterfactual substance without counterfactual definiteness*. Counterfactual discourse remains fully truth-apt—*would*, *might*, and *the probability that* receive verdicts by the semantics of section 2.2—while the assumption that all counterfactual measurement outcomes possess simultaneous determinate values, the assumption Bell-type theorems quantify over, fails *natively*: an agreeing superposition admits no alternative in which its attribute takes a single value, so the joint valuation the theorems require does not exist to be forbidden. This is precisely the combination the quantum realist program has needed: enough counterfactual structure to say what experiments would have shown, and no hidden valuation for the theorems to kill. Hidden-variable programs supply the first half of the demand at the cost of the second

(they say what would have shown, by carrying a valuation of the kind the theorems restrict, noncontextual and measurement-independent; a contextual, nonlocal program such as Bohmian mechanics survives them, as the framework acknowledges); instrumentalisms supply neither; the Aethic structure supplies both. The collection begins immediately, with the structural move that the note has just made safe to state at full ontological strength.

One question is answered here rather than left to fester: a classical filtration also retains its past $\sigma$-algebras, so the framework's retention of indices is not what distinguishes it from classical epistemology; what distinguishes it is that yesterday's openness is present content at yesterday's index, consulted by the counterfactual clauses, rather than a fact about yesterday reported from today.

# 4 Physical Disjunction, and the Union Move

The semantics so far treats one blank attribute at a time, and the note above has just argued what such an attribute *is*: open, co-present alternative content, permanently indexed. But that argument also creates a debt, and it must be paid immediately: if the alternatives are co-present—physically here, in the one block—why does any observer ever see just one? A framework that quietly borrowed mutual exclusivity back from classical metaphysics at this point would be undoing its own closure. The union move is the payment, and it is therefore not an optional stylistic commitment but the forced next step: now that co-presence is earned rather than posited, the account of non-interaction must be rebuilt from materials the framework actually owns—and it is the move that will carry the physics. Its one-sentence form, the framework's deepest inversion: exclusivity is an *achievement*, not a default — stated interior content that some configurations of alternatives carry and others simply lack.[4] Classical metaphysics treats mutual exclusivity as primitive: alternatives, whatever they are, exclude one another by nature, and *or* in its physical sense simply inherits that exclusion. The Aethic proposal inverts the inheritance. Enumerating a blank attribute's alternatives—the disjunction—is one job; *mutual exclusion* between the enumerated alternatives is a separate job; and the framework hands the second job to the *interiors* of the enumerated pockets, where it is done, when it is done, by stated content — and the content in question is a *logical declaration*, not a qualitative one. Where the interiors state the mutual negations — each alternative's cross content with its rivals declared *false*, side by side — the alternatives exclude: co-present but mutually barred, and this is the *disagreeing* superposition, the kind every example so far has involved, and the kind whose macroscopic ubiquity makes classical metaphysics feel true. But where *nothing is said* — the interiors silent, no negation declared, no affirmation either — the joint standing of the alternatives remains open, and this is the *agreeing* superposition. (The familiar gloss that agreeing alternatives "interact" is pedagogy, and it is retired here before it can mislead: the dichotomy is logic — declared-false against undeclared — and which class of physical interaction actually runs on an agreeing pair is fixed by the mechanics that consume the logic, not by the logic itself; in particular, quantum interference is not mechanical interaction wearing a mask.) As structure: with $e_{ij}$ the cross attribute carrying the joint standing of alternatives $i$ and $j$,

disagreeing: $\{a_1, \ldots, a_n\,;\, e_{ij}$ declared false$\}$ agreeing: $\{a_1, \ldots, a_n\,;\, e_{ij}$ undeclared$\}$,

[4] Two lineages, since this move is squarely a revision in the neighborhood of disjunction. Against quantum logic [11] and the empiricism about logic it prompted [33]: the revision there is to the *consequence relation* — a non-distributive lattice of propositions, connectives reinterpreted, deduction weakened — while here deduction stays classical and the propositional lattice distributive, and what changes is the *primitive stock*: physical union enters as a new composition primitive — the jointness of *or*, per the display's note below — rather than as a reinterpretation of the connectives. So Putnam's question does land on logic, and receives a changed answer: the empirical matter is not the consequence relation but the primitive inventory, and, downstream of it, which exclusions the world's records state — with the one distributivity question the framework does own priced as an explicit hypothesis of the state algebra (the unweighted semiring of the algebra companion), not of the logic [4]. At the grade a quantum logician will insist on, the differentia is *where* the revision lives—in the state algebra rather than in the consequence relation—and not whether there is one; the framework claims the location, and prices the algebra's distributivity accordingly: the algebra companion exhibits its failure outside restricted fragments, so it holds as a hypothesis on those fragments only [10].

one attribute's worth of difference, carrying the entire classical–quantum divide.[5] Written as algebra, the two forms are

$$\text{agreeing: } \bigcup_i a_i \qquad\qquad \text{disagreeing: } \bigcup_i \Big( a_i \cap \bigcap_{j \neq i} \neg a_j \Big),$$

where the lower-case letters are pegs of one attribute and each term of the disagreeing form is a state whose interior holds one peg and, filling the attribute with exactly that peg, excludes the rest, an interior being always of a state and holding pegs, never other states or their negations; and the placement of the exclusions is the entire content: the non-interactivity that makes the road sign appear on only one side lives in the *interior* of each pocket, never in an exterior wall between pockets. The two sums in play are one filling read under two intersected conditions: the pegs inside a filling are the attribute's open values and say nothing about exclusivity; intersected with the single-peg carving, the filling is the union of the determinate states, the disagreeing form above, its outer union composing states; intersected with the irreducibility marker, its determinate refinements are invalid and it is the agreeing superposition proper [4]. There is no physical "or" holding alternatives apart; there is only co-presence, with or without interior negations—which inverts the classical picture, since jointness remains the unobstructed default and mutual exclusivity the manufactured special case, interference being one physical use of that unobstructed jointness.

Jevons's inclusive *or* is the precedent for how a reform of this kind stays invisible: a change at the level of what disjunction is, adopted so completely that the prior reading becomes unthinkable within a generation. The precedent is for the reform's invisibility, not its warrant; the warrant is the removability of exclusion, and the comparisons with quantum logic and with the color-exclusion problem are the exclusivity paper's [4].

## 5 The Postulates, as Codification

The reader now holds an ontology, and it is time to say what its laws are. The framing matters: nothing in this section is a new request. The three postulates of the framework are codifications of rules the reader has been using since section 2.1, and the right way to read them is as an inventory of commitments already granted.

**P1.** *Indexing.* Every reality-attribution—every determinate *content*—is of the form "$q$ takes value $v$ relative to $A$"; there is no $A$-free ledger of content. What is un-indexed is *structure*: the validity functional and the rule from configurations to record partitions, the grammar over vantages that section 8 owns as such, which is not itself a vantage. A fact with no Aethus to carry it is not a modest fact but a category error, and the error has an exact shape: truth is not a property statement-objects carry on their own—a stated attribute taken alone is an unapplied thing, an operator dangling without an argument—and what truth *is*, categorically, is the event of application: the attribute meeting an Aethus's structure, and the structure responding.[6]

[5] The notation invites a classical misreading, and the concept beneath it is perhaps the framework's most alien. The two regimes are not two configurations of one classical substance — a list of alternatives with an interaction flag toggled on or off. The union principle beneath this display holds that composition itself comes in two logical substances: alongside classical conjunction there is a *primitive union* — the *jointness* of *or* — under which logical disjunction realizes physically as joint occupation of the block, co-presence with *openness* as its native character — a jointness usable but, like *and* itself, not further describable, it being a logical primitive. That substance is the *default* composition of co-present alternatives; classical mutual exclusivity is not primitive but manufactured, the nested achievement in which interior negations forbid interaction while permitting simultaneous presence. The familiar physical *or* is thereby, in the fuller taxonomy, entirely a subcategory of the disagreeing case — and the agreeing case is categorically alien to classical metaphysics, a sister taxon rather than a variant of anything within it. Logical realism survives the alienness: non-contradiction is preserved because disjoint attributes taken in *intersection* are exactly what the framework prunes as invalid. The blank $e_{ij}$ therefore marks no absent classical flag; it marks the primitive substance operating unimpeded, the stated $e_{ij}$ marking its manufactured suppression. And one division must not blur: *interference*-grade interaction is no property of this attribute-side substance — it is how *matter* uses the opened jointness, the quantitative layer filling paths into the void the attributes open; or-jointness is an abstract attribute-mediated state, and the interaction is its material use [4].

[6] The long paper's image for the same point: truth is one half of a torn fabric whose irreducible other half is the ancestor–descendant structure of a selected Aethus. "Actual" then receives a definition rather than a primitive role: the

**P2.** *Blanks.* Relative to any valid Aethus $A$: an attribute $q$ is in superposition relative to $A$ if and only if $q$ is blank relative to $A$—blank in the retrieval sense of section 2.1, its entire answer space open, which for an elementary attribute is exactly $A(q) = *$ and for a composite is stricter than $A(q) = *$ (a mixed composite is unstated without being in superposition). The mixed grade arises in this paper at exactly one place, a reopening's intermediate under a rival reading (section 2.3); in every use of the postulate below—the organizer's which-door attribute, the cross attribute $e_{ij}$, the powerset's bins, the entangled pair—the target is elementary or wholly blank, and $A(q) = *$ and blank coincide. Blankness *is* superposition, not a pointer to a hidden value—the bookkeeping rule of section 2.1 restated as a law, now that §3 has priced the deflation.[7]

**P3.** *Pruning.* An alternative survives only if it can be consistently refined, and alternatives failing this are not improbable but *invalid*: pruned, with no standing in the realizable residue, the tagged component remaining represented beneath its mark. Stated with full care the constraint is a two-quantifier $\exists\forall$ rule over the *proper-descendant* relation: writing $B \leq A$ for "$B$ is a *proper descendant* of $A$"—a *descendant*, resolving any set of $A$'s blanks along its own standing superposition structure ($B = A$ included), its cells terms of $A$'s decomposition at coefficients not measure-zero relative to the rest—the relation is reflexive and transitive, any number of resolutions compacting into one, a proper descendant of a proper descendant being one, and the rule's two quantifiers range over descendants at every depth, not over two rungs of a ladder, the proper-descendant relation being the subrelation of content inclusion $\sqsubseteq$ the postulate ranges over, as fixed among the Terms of §1. A refinement is the *selection* of an alternative that the graph already offers and affirmatively renders *resolvable*, and that the standing books do not forbid, never the arbitrary insertion of content from outside the structure.[8] Validity then satisfies

$$\mathcal{V}[A] \implies \exists\, B \leq A,\ \forall\, C \leq B,\ \mathcal{V}[C],$$

with $\mathcal{V}$ the *largest* predicate disjoint from the base contradictions and satisfying the displayed implication—the greatest fixed point of the safety operator, §5.3, since the implication alone is a necessary condition satisfied by the empty predicate—a $\exists\forall$ alternation whose necessity the long paper derives twice over. Read as a game: the observer refines, nature refines in reply, and $A$ is valid exactly when the observer has a refinement that survives every reply.

### 5.1 The Scope Split: Predetermination per Aethus, Randomness over Aethae

Two scope disciplines govern every claim the postulates make, and both are promoted here from the footnotes they previously occupied, because each is load-bearing enough to be pointed at by name. The first fixes what *predetermination* means and where it holds. Blanks apply to disagreeing as well as agreeing superpositions, and under P2 they are, by construction, constructive—literal reflections of a given Aethus's reality: to a given Aethus, the block universe it describes holds blanks, standing as superpositions, as literal content within it, the existence-state at each point strictly not undermined by any other point. Looking at a specific block universe, everything conspires under the constraint that it exists—fixing any point freezes it into that block, with the freeze propagating as a constraint

actual is the state of a stated attribute surviving the pruning (P3 below), when intersected with a given Aethus; the false, the state of that intersection being invalid. This is the commitment granted the moment the books were opened, which the note of section 3 argues cannot be walked back.

[7] The extrusion principle's deepest job is enabling this postulate, and the doctrine that job carries—predetermination per Aethus, randomness over Aethae—is promoted from this footnote to its own titled statement, §5.1.

[8] "Refinement" throughout this paper means exactly this relation, reflexivity included by design: the resolvability restriction—a refinement selects an alternative the graph affirmatively renders resolvable, as against a mere extension by arbitrary content—is what the long paper's "proper" marks, and it is not strictness; the trivial refinement $B = A$ is the formal counterpart of the waiting the derivation below invokes. The deeper characterization is the companion's reduced form, $\bar{A} = \sup\{\, B \leq A : \forall\, C \leq B,\ \mathcal{V}[C] \,\}$, where it exists: P3 requires every valid Aethus to admit a safe proper child, and where the reduced-form construction is established the surviving content admits the corresponding reduced representation; the algebra companion proves, separately, a real-weight maximum for weighted combinations in its specified setting and an order on a parent's partial fillings under stated representation assumptions; their correspondence with the general reduced form is open, so existence of a safe proper child is not by itself existence of the stated supremum. A terminal Aethus—no blanks left that the graph affirmatively renders resolvable—is classified by its own direct consistency, standing as its own reduced form through the reflexive self-refinement.

to everywhere else therein, the *pre-* naming the proper-time direction along which that consistency is read. The framework then places whole Aethae in disagreeing superposition, and every branch carries equally that much predetermined-ness, fixity of its rendered structure, blanks included, and not a value for every attribute; what Aethic pruning contributes empirically is not more determinism but more visibility, removing joint combinations so that per-branch consistency shows through. *Predetermination per Aethus, randomness over Aethae*: the two claims are never in tension, because they hold at different scopes, and conflating the scopes is precisely what makes the accounting look paradoxical. Delayed choice in Wheeler's family is the structure's special case—what stands in *disagreeing* superposition is the joint interference-with-no-detector case against the joint no-interference-with-detector case, one phenomenon made empirically obvious one joint branch at a time (the quantum-controlled family, where the configuration pockets stand agreeing, is sorted at section 7)—the agreeing superposition of paths standing between determinate nodes in any case, its character never overwritten anywhere in the block.

The second discipline is the split between *measurement* and *observation*: two processes, not one. Measurement is the transfer that asserts a record into the world's own sector—an interaction whose posterity any future index can inherit—while observation is the indexing transit by which a particular Aethus comes to state what the record fixed; the boundary theorem of section 7 classifies by the first, collapse in section 8 is the second, and most of the measurement problem's classic confusions are the two processes run together under one word (full development in the companion [4]). Everything downstream should be read against both disciplines, and the satellite papers point here when they invoke either.

### 5.2 Where the Third Postulate Comes From

P3 deserves its derivation on the page, because "collapse is derived" is this paper's sharpest claim and the derivation is short enough to audit. Run the double-slit experiment in a deliberately macroscopic costume: participants are secretly assigned one of two doors and a position to walk to; an organizer sits blindfolded until the end. The organizer's Aethus leaves the which-door attributes blank, so P2 places them in superposition—and yet it is guaranteed that, on removing the blindfold, the organizer sees two clusters and never an interference pattern. This, named, is Schrödinger's cat played macroscopically straight: the fate-attribute is blank relative to the outside Aethus, so the superposition stands by the second postulate alone—the semantics' residue operation (section 2.2), with quantum mechanics never consulted—and the macro-scale record supply prunes it to the *disagreeing* kind: exclusive branches, no interference to find. The companion prices the trade without flinching: ordinariness across the macroscopic world is sacrificed—everything classical becomes subtly stranger—to instate realism into quantum mechanics. Why the two clusters, then?

The naive answer—*a future version of the organizer could simply ask a participant, and the answer would contradict the interference pattern, so the interference state is impossible*—fails on inspection, and the failure is instructive. The contradiction occurs in a *future* Aethus, and P1 forbids exporting contradictions freely between Aethae: if a clash in any one information state could invalidate any other, implication would flow between arbitrary Aethae and an observer-independent background *content* would be smuggled back in through the plumbing. So the puzzle is sharp: the *possibility* of a future contradiction must constrain the present Aethus without violating the framework's own prohibition on arbitrary cross-Aethus traffic.

The derivation proceeds by elimination over the candidate rules. A single-quantifier existential rule—invalid if *some* refinement is invalid—is too strong, and the companion's inheritance diagram makes the failure exact: let a state $A$ hold refinements $B$ and $C$, with $C$ invalid. The rule strikes $A$ through $C$; but invalid inheritance—every refinement of an invalid Aethus is invalid, the one corollary that follows from the definition itself (immediate at the clean-future grade the companion's fixed-point lemma certifies)—then forces $B$ down with its newly condemned parent, while $B$ stands valid by construction, healthy refinements and all. One bad option among many condemns its innocent siblings, which is the collision stated directionally: invalidity flows from a state to its refinements, never from a refinement to the state [4]. A single-quantifier universal rule—invalid only if *all* refinements are contradictions—is too weak: the organizer's state always has refinements that are not yet contradictions, since one can simply

wait, so the rule never fires; worse, those waiting-states would have to ground each other's invalidity in a circle with no base case. What breaks the impasse is a structural observation: the problematic states are *self-similar*—every non-contradictory refinement of the organizer's state is another state of the same kind, facing the same inescapable menu. Folding that self-similarity into the existential fact that each such state has *some* contradictory refinement yields a two-quantifier alternation, and generalizing gives P3 exactly as displayed above. (Elimination over three candidates is elimination only if they exhaust the field, and the field is the unweighted quantifier-prefix rules on the refinement relation—fixed as such by the division of labor section 2.3 states, validity binary and weight graded, which excludes any rule keyed to the weight of contradiction-reaching refinements and therefore stands among the elimination's premises; P3's *invalid, not improbable* clause restates that division rather than supplying it. Within that field a rule is fixed by its first quantifier and its second, each ranging over refinements, which are descendants at every depth. At the first quantifier an existential is excluded by the inheritance direction—a rule that condemns a parent through one descendant condemns that descendant's innocent siblings, whatever they hold deeper—so that quantifier is universal or the rule is wrong; at the second a universal never fires, since waiting is always a refinement, so that quantifier is existential or the rule is idle. That leaves the alternation displayed, and it is already closed: read over descendants at every depth, "some refinement all of whose refinements are safe" reaches the whole cone in one application, so no deeper nesting adds anything it does not contain, (section 5.3).) The chess reading is then literal rather than decorative: *check* is the escapable possibility of contradiction; *checkmate* is the impossibility of escaping that possibility; and checkmate is invalidity. The postulate says: it must not be impossible to make contradiction impossible. The organizer's interference-Aethus is checkmated—every refinement leaves which-door information retrievable in principle, by asking, by records, by entropic traces—while the two-cluster alternative has a clean future and survives. Collapse, in this costume, is a theorem about which bookkeeping states have escape routes. The column discipline of section 6 binds this claim too. What the elimination derives is the *form*, the two-quantifier alternation; the base contradiction it runs on—that a refinement stating both the interference and a which-door value is invalid—is supplied by the union move's identification of interference with record-absence (section 4), a posit of the ontology rather than an output of the elimination, without which the organizer's interference-state has no contradictory refinement to find. "Collapse is derived" reads at that grade: form from the elimination, trigger from the union move—Feynman's rule formalized, which section 7 collects as a deliverable.

### 5.3 The Fixed-Point Form

P3 has a standard mathematical shape, and naming it is what plugs the collapse law into an existing, well-understood genre—though, as the paragraph ends by saying, the shape closes at once and nothing in it recurses. Collect the non-recursive failures—direct contradictions among stated attributes, P2 violations—into a set Base, and define the *doomed* Aethae as the least fixed point

$$\mathfrak{D} \;=\; \mu X.\big(\,\mathrm{Base} \cup \{\, A \;:\; \forall B \le A,\; \exists C \le B,\; C \in X \,\}\,\big),$$

so that an Aethus is doomed when every proper descendant has a doomed further one. Here $\le$ is *declared descent*, the splittings each Aethus admits, and not the lattice order, which would place the tagged all-statements Aethus beneath everything: an attribute whose single-peg carving the Aethus holds splits into its values, a filling asserted without one declares no split, and an irreducible agreeing superposition splits only when a record grants access, into children that are all tagged. Stated so, and with finitely many attributes and the algebra paper's closure rule, that a carrier holding a carving each of whose settings is tagged is itself tagged, doom coincides with closure's own tag and the doomed Aethae form a multiplicative ideal, so that validity is an exact quotient of the algebra [4]; beyond finitely many attributes the coincidence is open. Validity is the complement of $\mathfrak{D}$: the *greatest* fixed point of the dual safety operator. This is precisely the shape of a safety property in the modal $\mu$-calculus, and of the non-losing positions in a game against nature—which the chess idiom names, though as an idiom only: membership in $D$ says that every declared descendant has a tagged descendant, which is weaker than every continuation meeting a tag in finitely many steps, and finite branching supplies no finite certificate; finite termination holds separately in models with finitely many attributes, where each

declared split fixes an attribute for good. The Knaster–Tarski theorem guarantees both fixed points exist for arbitrary branching, so nothing here presupposes finiteness; and because declared descent is reflexive and transitive—a proper descendant of a proper descendant is one, any number of resolutions compacting into one—the iteration closes at its first stage for any branching whatever, so that P3's own $\exists\forall$ statement, read over descendants at every depth, is the fixed point rather than a seed for computing it: a state is valid exactly when some descendant has a cone free of base contradictions. The equivalence between P3 as stated and this fixed-point formulation, under the base-generation reading of invalidity, is stated and proven as a lemma in the long paper [4]; the short version's job is only to display the shape. The least fixed point also carries the framework's own recursive reading on its face, its quantifiers ranging over every declared depth at once, and, as noted just above, finite branching supplies no finite certificate: a finitely branching waiting chain with a contradictory exit at every stage has every position with access to contradiction while an infinite waiting continuation never takes the exit. Finite termination holds separately where each declared split fixes an attribute for good and the attributes are finitely many.

Validity and the *reduced form* are then the engine. The reduced form of an Aethus $A$ is defined as the supremum, by Aethic parentage, of the set of proper children of $A$ all of whose own proper children are valid—informally, the maximal safely-refinable core, the residue of alternatives surviving the pruning. Three results bear on it and are not to be run together: the order of a parent's partial fillings, under its representation assumptions; the maximum attained by real-weight combinations, their valid-support truncation; and the correspondence of that maximum with this supremum, which the algebra companion books as open [10]. P3 requires every valid Aethus to admit a safe proper child; where the reduced-form construction is established, that is the statement that the surviving content admits its reduced representation. Two facts give the definition its standing. First, an equivalence: by the Aethic Partition Theorem of the long paper, the union of all elements of a partition equals the union of its valid elements alone, since invalid Aethae contribute zero weight (weight-zero being the algebraic face of the invalidity label, which the term keeps, §10)—so the reduced form is Aethically indistinguishable from the original at the tier that matters, agreement of realizable content. Pruning discards nothing that was ever available to be inhabited. Second, every operative notion of the paper has already been defined over it: a counterfactual's verdict is read off it, a dependence is checked by whether one perturbation changes it, and weight lives on it. One property, before the derivation: the pruning is driven by *stated information*—records, correlations, anything an Aethus carries—and by nothing else. Not size, not mass, not the presence of anyone's gaze. That austerity is about to do the work.

## 6 One Operation, Many Phenomena: The Reduced-Form Coupling

When the pruning eliminates *joint* configurations of several attributes, the survivors exhibit a correlation that nobody put in by hand: certain values of one attribute now only ever occur alongside certain values of another, not because anything signaled anything, but because the combinations that would have broken the correlation were crossed out. Call this *reduced form coupling*. The framework's compression claim—that phenomena standardly treated as unrelated are one mechanism—rests on this operation, so it should be watched running on the humblest possible case before it is claimed for entanglement. Every step below is the same operation later run under quantum weights.

Take the Monty Hall problem, in one paragraph, since its result is placement and not evidence. Write $B_i$ for *the car is behind door $i$* and $O_j$ for *the host opens door $j$*; the uniform prior is the weighted Aethus $A_{\text{orig}} = \frac{1}{3}B_1 + \frac{1}{3}B_2 + \frac{1}{3}B_3$, the contestant committed to Door 1 by convention. The host's conditional-response rule is itself an Aethic object—the *Monty Hall Nexus*, to this problem what the Quantum Nexus of section 9 is to the physics—and with the pick stated its survivor is $X = \frac{1}{2}(O_2 + O_3)B_1 + O_3B_2 + O_2B_3$, best read by its *absences*: $O_1$ appears nowhere (the host never opens the picked door); $O_2B_2$ and $O_3B_3$ appear nowhere (never the car door). Those joint configurations are Aethic contradictions that never enter the structure—the host's "choice" realized as an invalidation mechanism, not a dynamical influence—and the lone $\frac{1}{2}$ marks the one branch where the host retains an option. Intersecting, $A_{\text{after}} = A_{\text{orig}}X = \frac{1}{6}B_1(O_2+O_3)+\frac{1}{3}B_2O_3+\frac{1}{3}B_3O_2$, (using $B_iB_j = 0^{\times}$ for $i \neq j$—the *marked* zero, an invalid intersection carrying weight zero beneath its mark, the closing ledger's design

choice—and idempotence, the composition primitive's, the union's $X + X = X$): four pockets, weights summing to one. The host opens Door 3, one more stated attribute: every $O_2$ pocket is invalid against it, the residue $\frac{1}{6}B_1O_3 + \frac{1}{3}B_2O_3$ has total weight $\frac{1}{2}$—gone from the residue, the crossed-out pockets standing as marked terms with their weights intact beneath the mark, which is what a later reopening recovers, §10—and rescaling gives $\frac{1}{3}B_1O_3 + \frac{2}{3}B_2O_3$. The familiar two-thirds is not produced by a conditioning formula; it is the residue of crossed-out joint configurations, their weight flowing to the survivors, and the long paper proves the general fact: Bayesian conditioning *is* pruning-plus-renormalization, an identity of the base rather than an added rule. A column discipline applies from this first instance: results of this kind (conditioning here; no-signaling and the Adams alignment later), are *placement* results, definitional once the base is fixed, while the evidence column is occupied elsewhere—the classification theorem, the non-preferentiality result, the staked divergences of section 8, and the two-tag geometry of section 9.1. The "where the car is" and "which door opened" attributes are coupled by validity-preserving pruning alone, with no causal mechanism invoked anywhere.

A larger prize rides on this than the game show suggests. Weight, in this framework, is *ontological*—a structural property of the tree, with probability as its formal definition—and it is indexed by construction: there is no $A$-free ledger, so every probability attribution has an Aethic reference state, written or suppressed in notation; marginalizing within a fixed index introduces no observer-free vantage and is permitted. One object thereby underwrites what the philosophy of probability has spent generations trying to unify. Single-case chance and long-run frequency are the same weight read once at a node and once along many, under the independence or convergence conditions the laws of large numbers require; credence-updating is the pruning just displayed; quantum probability is the same weight under the amplitude kernel of the next sections—four readings, one structure. Its realization is centric unfolding, the agent's occupation of successive Aethae, which the long paper treats as one kind of resolution for quantum and classical attributes alike, its Randomness Universality Principle [4]: the amplitude kernel supplies the numerical weights for quantum attributes, not a separate kind of randomness. Centric unfolding prescribes the eventual outcome probabilities of a specified resolution, its weights in the validity quotient, while leaving the finite-step process underdetermined within its admissible family. And the account is detachable: it stands on part one's machinery alone, so a reader who suspends every quantum claim in this paper can adopt the probability ontology by itself.

Now vary only the weights. Under amplitude weighting the identical eliminate-the-joint-configurations operation presents as quantum entanglement: Bell correlations are reduced-form coupling carrying the quantum weights, the surviving joint outcomes exactly those no future Aethus can catch in a contradiction—and run across observers, the same coupling enforces cross-observer consistency, every contradiction-pocket pruned by P3, so the observers who can meet are the ones who agree. Wigner's friend and Frauchiger–Renner receive their full, graded treatment at the nested-observers entry of section 8, which this coupling feeds. One operation; four literatures. The methodological point is the core of the compression claim: the reduced form was not engineered to unify these. It is one deductive step from the postulates—P3 stated positively—and the unification fell out afterward, which is at least mild evidence that it tracks structure rather than bookkeeping convenience. The long paper carries this engagement one step further into the causation literature, where the Wood–Spekkens fine-tuning result shows that classical causal explanations of Bell correlations must be fine-tuned, a burden the absence of causal arrows does not by itself discharge: reduced form coupling is offered there as that substrate—neither signal, nor common cause, nor retrocause, but pruning—and the engagement carries two grades that are distinct: in disagreeing superposition Reichenbachian *screening* is a theorem—pairwise-composed couplings, $W(a, b, c) = f(a, c)\, g(b, c)$, give conditional independence at every positive-weight pocket by a one-line factorization—and the rest of Reichenbach's principle is not derived: the causality companion supplies connectivity, separation, screening and covariance identities for factorized states, with screenability relative to the attribute stock and causal common ancestry not among the derived results.

# 7 The Climax: The Boundary Is Informational

Take a system with enumerable alternatives—the laboratory case will be a particle with available paths—and instead of considering the alternatives one at a time, consider the *powerset*: every combination of alternatives held together, each combination a candidate pocket of co-present content. The framework's constraints now act on the powerset, and the theorem that organizes everything downstream is this:

Fix the laboratory case: $n$ slits and an arbitrary configuration of detectors, each detector firing on traversal of any slit in its designated subset. Everything that follows is indexed, per P1: the configuration is stated content relative to the Aethus $A$ doing the classifying, and the partition and the bins are $A$'s. Where detector-presence itself stands blank relative to $A$, the classification runs pocket-wise over $A$'s configuration pockets, one partition per pocket, and the criterion of section 4 sorts the delayed-choice literature into its two families. Where the setting was chosen and recorded and only $A$ has not learned it, the pockets are *disagreeing*: one partition obtains, unread—Wheeler's case [38]. Where the setting is carried by a quantum control uncorrelated with any path record [26], the pockets are *agreeing*: the partition itself stands in agreeing superposition, and no fact of which obtains is available to be unread—the wave–particle intermediates those experiments report. How the cross-partition terms are carried is the weighting's business, stated at §9.1; the family is previewed here only so that it is seen to be handled, and the theorem uses none of that machinery. What is *not* indexed is the rule from a stated configuration to its partition, the shared grammar of every vantage, which section 8 owns as such. Throughout, a *detector* is a device whose marker states for distinct slits are orthogonal, so that its signature is deterministic; overlapping markers are the kernel's business (§9.1). Assign each slit its *detector signature*—the tuple recording which detectors would fire if that slit alone were traversed. Slits with identical signatures are indistinguishable by every record the signature includes; that no record, present or future, separates them is the completeness of the signature for the declared configuration, stipulated below, and that the separation persists through every proper descendant is a second stipulation, persistence, distinct from the first. The signatures therefore partition the slits into equivalence classes $E_1, \ldots, E_k$, and the theorem classifies every element of the powerset against that partition.

**Theorem 1** (Powerset Invalidation)**.** *Assume the declared record structure is complete and persistent: the signatures include all relevant accessible records, and every proper descendant retains a descendant stating a separating record whenever the candidate spans signature classes. Then every subset $T$ of the $n$ slits—every candidate claim "the particle traverses exactly these slits in coherent superposition"—falls, relative to an Aethus in which the detector configuration is stated and over the specified candidate domain, into exactly one of three bins: (i)* valid*: $T$ is empty or equals one full equivalence class; (ii)* over-specified*, invalid by record-individuation: $T$ is a nonempty proper subset of a single class, drawing a distinction that no information could ever support; (iii)* third-postulate invalid*: $T$ spans two or more classes, combining alternatives some record can distinguish, so that a future refinement can expose the maintained coherence as contradictory—checkmate, in the sense of section 5.3. The bins are mutually exclusive and jointly exhaustive, with the counts explicit: $k$ valid nonempty cases, $\sum_j (2^{|E_j|} - 2)$ over-specified cases, and the remainder third-postulate.*

*Proof.* For nonempty $T$, the signature partition gives an elementary trichotomy: $T$ is a proper subset of one class; $T$ equals one class; $T$ meets two or more classes. In the first case $T$ distinguishes slits that share every possible record, so by record-individuation no attribute carries the distinction and, over the specified candidate domain, whose preparation fixes the classes, the claim over-specifies. The bin rests on record-individuation and that domain, not on P2 alone: the framework recognizes mixed, partially constrained attributes, so a proper part of a class is not excluded merely for falling short of a whole answer space; the companion's preparation-relative classification is the version followed here [9]. In the second, every branch of the maintained superposition produces the identical total signature, so no refinement along accessible records can expose a contradiction, and the state has a clean future—valid (empty $T$ trivially so; with no detectors, $k = 1$ and the sole valid nonempty case is the full set). In the third, some detector's record separates a slit of $T$ in one class from a slit in another, so the coherence claim conjoined with that record's future stating is contradictory while the record family is persistent over

the candidate's declared descent, every refinement having a further descendant stating the separating record, which is the persistence hypothesis and not completeness, completeness fixing only what the signature includes—checkmate in the fixed-point sense above, third-postulate invalid. Exclusivity and exhaustiveness are the trichotomy's own, and the counts follow by enumeration. □

Children, fixed: what counts as a refinement of the classifying state is determined by the configuration that state itself states, along its signature partition—so in the no-detector configuration the singleton resolutions inside a class are not refinements of any state, and P3, which quantifies over refinements, never reaches them; the axiomatization companion states this as its Lemma (Children in a configuration), and without it the paradigm valid state would be doomed by its own bin-(ii) singletons [9].

A corollary worth one sentence: moving or erasing a detector re-draws the signature partition and re-bins the powerset—the sliding quantum–classical boundary as literal re-classification. And the theorem's shape in one breath: bookkeeping over a partition plus two postulates, with everything quantum entering through which partition the records induce.

Since a theorem asked to explain this much will have its premises inspected aggressively, here is the complete inventory: the three postulates, the enumerated-alternatives setup, a complete and persistent record structure whose cells are structurally visible, and the imported record-individuation principle booked below. No dynamics, no Hilbert space, no quantitative kernel weights and no scale parameter enter: those arrive with the weighting axiom, and the classification's independence is from the quantitative kernel, not from the structural visibility proper childhood uses. Two further guards on the setup, both owed to a careful reader. Signatures quantify over *all* correlated degrees of freedom, preparation included: aim a beam at one slit with no detector anywhere, and the collimator itself is the record that licenses the singleton—without that totality the theorem would misclassify ordinary preparations. A record, throughout, is not a physical primitive but the framework's own notion—any degree of freedom whose state is correlated with an alternative and is stated in some refinement of some Aethus, the record-individuation lemma read as a definition, with the long paper's Fundamental Theorem stating the same individuation from the other direction—a statement about distinctions, not values: where no record can ever be taken there is no distinction to draw, so nothing selects and the alternatives are performed jointly, which supplies no value and only fixes the grain of the attribute space for that index; what the framework does not supply is the dynamics that make degrees of freedom correlate, so every boundary statement is exact conditional on the physically realized record structure. One further principle earns its booking here, since the inventory above must stay honest: classifying a proper *subset* of a signature class as over-specified uses *record-individuation*—alternatives are individuated by possible records, so a distinction no information could ever support is not a distinction the attribute space contains. It is derived in the companion (its Record-Individuation Lemma: an indiscernibility principle for attributes—no possible record, no question—from the definition of an attribute and P1 given the record discipline, with permanence blocking "determinate but unaskable" [4]), and stands here as imported rather than smuggled, the record discipline it runs on being the booked item. The totality invites its own worry, which is better met than dodged: if signatures are total, how much of the theorem is bookkeeping? Exactly the part that could have been otherwise is not: the three-bin exclusivity, the two limiting cases, and the discreteness of the boundary are classifications a *different postulate set* would fail—content as a consistency check the theorem passes, not as empirical contact, which section 9 assigns elsewhere. And the third bin's trigger faces a known dilemma: under strict global unitarity nothing is absolutely irretrievable (the third bin then swallows everything; it is the valid class that threatens emptiness in a closed universe), while for-all-practical-purposes retrievability would make collapse approximate (inheriting the standard FAPP critique). The framework's answer has a named type—accessibility relative to the standing record structure, with the recombination caveat—no erasure window left open, the delayed-choice discipline of this section's eraser reading—as its guard, that discipline being that an eraser never un-records: a state that carried a record carries it permanently at its own index, and what erasure produces is a later state lacking an accessible distinguishing record, so that "erased" names a downstream lack and never a predecessor's loss—and the exact macroscopic conditions are the long paper's fifth open construction, flagged there rather than papered over [4].

The theorem's interest is not its difficulty but its placement: under the stated structural-visibility assumptions the boundary is drawn before the quantitative path-kernel weights are assigned, by the records available, and it moves when a detector is added, moved, degraded or erased, which is the sliding boundary the double-slit experiment reads off directly.

The double-slit experiment is then not an interpretive puzzle but a one-page reading of the theorem. With no detector at the slits, the Aethus of the experimenter carries no record fixing the particle's path; the both-slits element of the powerset survives as an agreeing superposition; and by section 4, co-present alternatives with blank interior non-interaction *interact*—the receipt issued there is collected here, because the interference pattern on the screen is what that interaction looks like when the alternatives are paths. Switch the detector on, and a record now exists: the both-slits element's interior non-interaction is stated, the element is invalidated, the surviving pockets are the one-slit alternatives in disagreeing superposition, and the screen shows two lumps. Degrade the detector—partial which-path information—and the boundary slides continuously, fringe visibility falling as the stating spreads through the powerset, which is the experimentally familiar complementarity trade-off read as bookkeeping [34, 14]. Decoherence is the same sentence at environmental scale: the environment is a prolific keeper of records, so macroscopic combinations get their interiors stated wholesale, and the classical world is the powerset pruned to its disagreeing remainder. Nothing in any of this consulted a size, a mass, or an observer's consciousness; a detector whose record is erased before any correlation escapes restores the blank, and delayed choice is the theorem's indexing clause above read at both its settings (Wheeler's family [38] and the quantum-controlled one), since blankness was never dated by proper time in the first place.

The sliding boundary has a quantitative face, and stating it prices the framework's claims exactly. Mark the paths with a partial tag, and two regimes appear. Classically random tagging—a disagreeing apparatus superposition—yields fringe visibility $V$ and which-path distinguishability $D$ on the linear chord $V + D = 1$, which the framework *deduces*. The notation is fixed once, since it has confused several readings: $D$ is one quantity throughout, the which-path distinguishability accessible from the record at the placement in force—the total-variation gap between the two outcome distributions the paths induce on that record—and its supremum over admissible placements is written $D^\star$, which for pure markers is Englert's $\sqrt{1-c^2}$; the port predictability $P_\theta$ is not $D$, and the doomed set $\mathfrak{D}$ of §5.3 is not a number. The chord and the circle are statements about the same $D$; the circle $V^2 + D^2 \leq 1$ holds at every placement, with equality for pure markers exactly at the optimal one, where $D = D^\star$ (the two-line deduction is run at section 9.1, where the two curves are taught in order). Coherent partial tagging—one agreeing joint cluster—reaches the circle $V^2 + D^2 \leq 1$, with equality $V^2 + (D^\star)^2 = 1$ for pure tagging at the optimal placement, [23, 17], whose geometry section 9.1 runs, with its one in-formalism definition standing between the geometry and the law. The empirical bite: two setups on these curves can share every intensity pattern and differ only in what *could have been* extracted—the agreeing/disagreeing distinction meeting experiment on the one quantity that lives on unperformed measurements.

# 8 Collecting the Realist Program

Three collections, each brief because the machinery has already done the work.

Peres's dictum that unperformed experiments have no results [31], long a slogan in search of a semantics, acquires literal truth conditions on this semantics: the would-counterfactual about an unperformed measurement whose outcome the perturbation leaves open is *false*, not gap-valued, while the mights stay true and the grade stays defined—truth-aptness all the way down, with no single result for the experiment to have had.

The proper/improper mixture distinction (the classic "and/or" pressure point against decoherence-style accounts) receives a complete treatment on the same vocabulary: an improper mixture is the reduced description of a composite that carries the coherence, its alternatives jointly open only at the level of the composite and exclusive relative to the record-conditioned subsystem, equal reduced density matrices being locally indistinguishable, proper mixtures are disagreeing pockets each carrying its fact, the two share a reduced density operator and are indistinguishable on the subsystem alone, separating

in their eraser statistics only where the partner system's records are in hand; the passage between them is lawful, with the third postulate's accessibility trigger as the stated converter, and that same trigger fixes *which* exclusive decomposition a proper mixture carries—the density operator admits infinitely many, and the framework's answer is the one the actual preparation and record structure distinguishes [4]. The agreeing superposition is the framework's one nonclassical ontological posit, and everything realist hangs on its status. It is *real*: not an epistemic shorthand for ignorance, but standing structure in the block, indexed. It is *interior*: alternatives as co-present contents of one spacetime, never a plurality of spacetimes. It is *weighted*, with the weights read off the kernel rather than assigned. It is *located*, as the powerset theorem's surviving bin. And it is *fingerprinted*: interference and eraser statistics are what its presence looks like in data, which is why the framework can afford realism where instrumentalism goes quiet.

*The Bell correspondence, in full.* The Bell correspondence can then be stated in full rather than in passing: the joint valuations whose nonexistence Bell's theorem forces are precisely the blanks of a standing agreeing superposition—the theorem's missing values and the framework's open attributes are one and the same object under two descriptions, which is why counterfactual substance without counterfactual definiteness is not a slogan here but the structure's native condition.

*New, or synthesis.* The framework is a synthesis at the level of its ingredients and new at the level of its engine. Relational quantum mechanics, QBism, the epistemic-restriction program, and Fine's fragmentalism each supply a piece — observer-relative state assignment, the epistemic reading of the wavefunction, the constraint on joint knowledge, the metaphysics of standpoints — and none supplies the consistency law that says which indexed states may stand. That law is the third postulate, and it is what the companion's placement section argues is the framework's own contribution [7].

*Nested observers.* Wigner's friend [39] and its Frauchiger–Renner sharpening [20] are the cases where indexed determinacy does its most conspicuous work: the friend's outcome is determinate relative to the friend's state and open relative to Wigner's, and no further fact adjudicates between them because there is no un-indexed standpoint for such a fact to occupy. The Frauchiger–Renner protocol is the one case the framework grades as conditional rather than full, since it turns on whether an observer-grade record can be kept retrievable and erasable in the way the protocol requires — which is the open physical construction of section 10. The departure from the standard picture is exact and belongs here. Sealing the laboratory in the moment does not secure an agreeing superposition relative to the outside observer: P3 prunes on the availability of the record over the whole proper-time future, and tense-free blankness reads a future retrieval as a present one, so, under the no-safe-refinement premise, the laboratory's outcome alternatives stand in disagreeing superposition relative to the outside observer even before the record is read—exclusive, classical disagreeing superposition from the moment the friend measures, and passes to a single state relative to the onlooker's books when the record crosses, by selection along already-exclusive structure. Agreeing superpositions do not collapse; the naive "interior in quantum superposition until the door opens" is replaced by "interior in disagreeing superposition until the record arrives," and the protocol's coherent measurement across the record basis remains performable, and what the third postulate excludes, on the record claim, is the coherent preparation it presupposes; applied to the disagreeing mixture the framework assigns, the measurement yields the mixture's statistics, cross terms absent, so the coherent state assignment used to obtain the protocol's certainty inferences is unavailable under the stated record premise; the inference that fails is the certainty step, a superobserver's ok outcome fixing a friend's outcome, and the long paper computes the two four-outcome distributions, $(\frac{1}{12}, \frac{1}{12}, \frac{1}{12}, \frac{3}{4})$ coherently against $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ under the mixture. What makes the protocol's contradiction look assemblable at all is a conflation the fuller development names and dissolves at the root, of *measurement* with *observation*: decoherence does not end superposition but converts joint openness into exclusive openness with the weights called before the assembly rather than after (the framework's Sifting Principle), so "the friend has measured" fixes an exclusive entry relative to the friend, while the outcome need not be stated relative to the superobserver even though the accessibility of its record can already constrain the superposition regime admissible at that index [4]. Both are worked at length in the companion [7], along with delayed choice, the cat, EPR, and the

mixtures; the present paper states the mechanism and leaves the case-by-case to it.

The verdict's one dependency is a physical claim, not a parameter. The third postulate's test has the quantifier shape $\exists B \leq A\ \forall C \leq B$ on the validity side, so a candidate agreeing superposition is excluded only when every permitted refinement still has a contradiction-reaching descendant: one possible later record excludes nothing, and present sealing of a laboratory secures nothing either way. Frauchiger–Renner's reversal of the friend's measurement is exactly a proposed escape, and what would forbid it is that an observer-grade record structure exports its records unblockably, so that no refinement stands in which they are all gone. Three statements differ, and the exclusion needs the third: that the friend can communicate the outcome; that the specified experiment retains an avenue for later communication; and that every permitted refinement retains some avenue to a separating record, so that no safe coherent reversal exists. The third is a claim about the physics of records, stated here as a premise; it is the fifth open construction, the conditions under which distinguishing information is permanently hidden, or cannot be. Two disciplines keep it from going verbal. "Unblockable" must not be cashed as what unitary dynamics already calls irreversible decoherence, on which reading the claim glosses a shared prediction; it must concern records unitarity still predicts recoverable in principle, the spin-echo and Loschmidt-echo regime. And the interferometry program [3, 18] can only fail to find the point at which the escape closes, so the claim stands as a commitment to a prediction rather than a prediction (§9).

Given the premise, the detector and the friend instantiate one criterion: the same record–coherence incompatibility supplies the base contradiction for a maintained two-slit superposition with an accessible which-path record and for a maintained superposition of the friend's outcomes with an accessible outcome record, and neither exclusion requires that anyone has read the record. The scope is the whole laboratory, the composite proposed as coherently controlled, since a path subsystem may lack interference while the system–marker composite retains coherence, and a physical realization must specify the candidate configurations, their proper descendants and the record incompatibilities on which $P_3$ operates. Without the premise, the indexing reconciles the two descriptions and excludes nothing physical. What the framework shares with the dynamical-collapse programs is the position that coherence ends somewhere for observer-grade systems; what it does not share is a parameter.

The bomb tester [16] is a further application of the same full-future criterion. In the ideal Elitzur–Vaidman arrangement a dark-port click identifies a live absorber without an explosion in that run, and the Aethic reading places the result in the admissible weighted residue of the apparatus rather than in a transaction with an unrealized explosion. The response being tested is indexed to the trial event, and later evidence can settle that historical attribute; this evidence need not record the photon's path, since "record" names an emergent realization of the joint-invalidity structure. The argument shows how accessible bomb-status information, together with the response law and the candidate agreeing content, two surviving amplitudes recombining, yields a contradiction that no safe refinement escapes, the no-safe-refinement premise here being the persistence of a macroscopic bomb's explosion and intactness records. The live case has explosion, bright-port and dark-port weights of one half, one quarter and one quarter, the transparent dud unit bright-port weight; the weights come from the kernel and the response law, and the criterion decides the composition law they fall under.

*Coherent device.* The coherent-device case, a live/dud response held in agreeing superposition, is a separate claim from the ordinary benchmark, whose probabilities remain standard; it is decided in the long paper on the definition of the would-clause, on which *would* $a$ holds when the weight of valid pockets that declare $a$ false is zero and an agreeing state containing $a$ meets that condition as a stated $a$ does, so that the coherently fired photon engages the live response, which the bomb, as a valid detector, renders permanently accessible, and the Aethus assuming the status agreeing has no safe refinement: a proposed divergence from universal unitarity, with the device's bright-conditioned status predicted a mixture where the unitary model keeps it coherent, which is its operational test.

*The modern test literature, engaged.* The theorems built for this family of views — Pusey–Barrett–Rudolph, the extended Wigner's-friend no-go theorems, and the ontological-models framework they presuppose — are met in the companion [7], where each is shown either to quantify over a category

the framework does not contain or to price a premise it has already surrendered. The verdicts are not reproduced here.

*The measurement problem itself.* Collapse, on this framework, is an indexing transition—the answer to *which block universe is yours* changing as information arrives—and never a physical event within spacetime. No preferred foliation is consulted, because nothing happens *on a slice*: the collision between Copenhagen collapse and relativity, in which two admissible foliations disagree about whether collapse has occurred, does not arise for a transition that is not in spacetime at all. On the EPR pair [15] the story is short. When Bob measures, even "earlier" in some frames, the update is Bob's alone; relative to Alice's standing Aethus his outcome stands in superposition, and stands so permanently. When Alice measures, her index moves to an Aethus relative to which Bob's measurement always yielded the correlated outcome. Nothing travels; an address changes hands. Read operationally, the no-signaling identity holds, the kernel supplying the far-wing balance it rests on—the Jarrett–Shimony conditions proper being stated on an underlying model state that this framework does not posit—and outcome independence fails together with the hidden valuation it presupposed, Bell factorizability having been surrendered at section 3. The formal problem is thereby closed—no projection axiom is left standing—while the bridge from formal invalidity to physical unrealizability remains the open construction of section 10. The case-by-case treatment, including the three distinct locality claims the phrase "nothing travels" bundles, is the companion's [7].

Three named placements complete the map, at a sentence each. The Pusey–Barrett–Rudolph theorem [32] does not bite, because it quantifies over un-indexed ontic-state backgrounds of exactly the kind P1 refuses to posit: relative to a single Aethus assignments are unique, and across Aethae the differences are real but never co-located, so no one ontic state is co-described by two overlapping distributions—the epistemic *flavor* that motivated $\psi$-epistemic models is earned without the commitment the theorem executes. With the Spekkens negativity–contextuality equivalence [36] the framework *complies* rather than collides: the compliance rides on the refusal of a single sample space, not on the signature of the tree's weights (the companion admits signed weights at the weight level while keeping the term algebra inverse-free, so that no combination of weights ever annihilates a label; the classical regime stays non-negative exactly where the Kolmogorov results require it [4]), and the signed structure quantum statistics require is carried by the interference kernel, whose cross-terms are constituents of one squared modulus—which is also where the contextual hidden-variable constructions [27] deferred from section 3 are met. And on the records side, Quantum Darwinism's prolific-broadcast thesis [40] is, here, the record supply itself, adopted as mechanism with its delivered job priced at section 7, while Spekkens's toy theory [35] stands as the closest existing demonstration that informational restriction alone generates interference-adjacent phenomenology.

What the collection amounts to ontologically is one block universe rendered relative to each Aethus, the renderings refining one another along an observer's chain, with collapse a change of rendering and not an event; the long paper develops the rendering and its relation to the moving present [4].

## 9 The Quantum Nexus, Briefly

Everything above ran on the three postulates and enumeration alone. The quantitative layer enters through exactly one further axiom, and it should be met before the close: what it is, and what it already buys. The *Quantum Nexus* is the weighting: the Feynman kernel installed as the weight structure over the powerset of paths, interference carried by the kernel's cross-terms and never by any fraction of validity. And the granularity is exhibited by the expression itself rather than asserted beside it.

*The object is universal.* The object doing the work is universal, not case-by-case: the Quantum Nexus $N$ is itself an Aethus—one weighted object, built once, over the *entire* powerset $\mathcal{P}(X)$ of all possible paths—given here in a factored form of the companion's boxed one-integrand axiom [4],

$$W_x(T) \;:=\; \Big|\sum_{p\in T,\; x(p)=x} \mathcal{K}(p)\Big|^2, \qquad W(T) \;:=\; \sum_x W_x(T) \;=\; \sum_x \sum_{p,q\in T:\; x(p)=x(q)=x} \cos\!\Big(\tfrac{S[p]-S[q]}{\hbar}\Big),$$

$$N \;\propto\; \int_{\mathcal{P}(X)} W(T)\, A(T)\; d\nu(T),$$

where $A(T)$ is the Aethus in which the paths of $T$ run in agreeing superposition, the pair sum runs over paths of $T$ sharing a configuration so the whole-cluster weight $W(T)$ pairs paths within one element only, and $d\nu$ is the powerset measure:[9] the Nexus reads, in the notation itself, as weights multiplying Aethus-terms over the entire powerset. A system's valid weighted outcomes are obtained by *intersection with N*—never inserted by hand. Element-wise, that intersection delivers each surviving cluster weighted whole, the residue carrying the elements multiplicatively:

$$\text{residue} \;=\; \sum_{T \text{ valid}} W(T)\, T,$$

whose every cross-term pairs paths *within the same element* at a shared configuration—one number, attached to the one term. Here $\mathcal{K}(p) := e^{iS[p]/\hbar}$ is the pure-phase Feynman kernel, so $W_x(T)$ as the squared modulus and the cosine pairing above are one identity rather than a proportionality: the coherent square written out, its imaginary part vanishing under $p \leftrightarrow q$ path exchange. The identity fixes the kernel's face of the negativity discussion below: the cosine cross-terms are constituents of one squared modulus, so a cluster's weight is a non-negative number carried whole—the compliance itself riding not on that but on the refusal of a single sample space, section 8. Nothing in the formalism offers a finer-grained bearer of weight, so reweighting above the agreeing-superposition's resolution has no slot to occupy; validity acts on whole elements, a cluster surviving with its interior cross-terms intact as constituents of its single weight, or dying entire.

*The axiom's status.* It is the framework's single flagged axiomatic input, and it is *separable*: remove it and the ontology of parts one and two stands; install it and the numbers arrive. And its installation is the third site of the one relocation the whole paper performs. Part one relocated determinacy, from a one-place property of facts to a two-place relation to an Aethus; the union move relocated exclusion, from the logic of *or* to the records that write it; the kernel relocates probability, from a quantity derived by squaring an amplitude back to the root—the weight law is the primitive, installed as the Born rule's expansion written out in full, and the amplitude is what the weights factor into, recovered by polarization and held as an object by nothing, (footnote above). A reader who measures the framework by whether the equations changed will find that they did not, and will have measured the wrong thing: what the equations rest on changed, at three sites at once, and that is the Hilbert-space premise unveiled rather than exceeded. The axiom's full signature belongs at this display, so that *one* is an audit rather than a headline: its configuration space is the augmented-path space — each element a spacetime route carrying an internal route on the two-node state space, the Bloch sphere run as software on Aethic hardware — its action functional includes the internal sector alongside the spacetime one, and its measure ranges over both; three declared components, one axiom, the internal kinematics carried here and nowhere in the postulates, with the spinor-ontology question already flagged attaching to this signature where it belongs.

*What is assumed.* With the kernel installed the stated axiom package is closed: three postulates, the union move, one weight law, and, stated as assumptions where they are used, record individuation, the composition law with its posited unitarity, the calibration of the weight measure, product structure, and the physical bridge from signatures to records. The count is short because several clauses are booked under one axiom, not because their content is absent. No separate measurement or projection axiom is posited, and none is eliminated either: record structure, record-compatible restriction, calibration and the physical bridge are assumed instead.

*What is represented or derived under those assumptions.* On the finite path-kernel domain the

[9] Because the kernel is installed as the pair sum, the Born rule's expansion written out in full with its cross-terms, no probability amplitude is ever an object any Aethus holds. No state-vector amplitude is primitive content of an Aethus; the kernel does assign each path a complex phase, $e^{iS[p]/\hbar}$, as an ingredient of the pair weight, and the state vector is bookkeeping recovered from the weights by polarization, at each configuration by the axiomatization companion's lemma and across configurations only as far as its stated global claim reaches, three obligations of which are listed there and not yet discharged [9]. The weight law is the entire quantitative input, and amplitudes are what it factors into when a Hilbert space is wanted.

package recovers the specified measurement architecture of the standard formalism, together with a Hilbert representation of its cluster states, under the assignment $T \mapsto \sum_{p \in T} \mathcal{K}(p)|x(p)\rangle$ over the readout configurations $x(p)$, paths sharing a configuration adding coherently and paths at different configurations orthogonal, which is the two regimes at kernel grade. Valid agreeing clusters of nonzero weight are represented by rays; the agreeing union is vector addition, the superposition principle at coefficients $(1, 1)$; the signature partitions of records yield self-adjoint observables, every record construction being an operator; the weight of a record-restricted cluster is the Born rule; pruning-plus-renormalization is the projection postulate, as the Lüders conditioning component of an ideal projective measurement; products of path spaces are tensor products; the kernel's composition law, with unitarity posited as part of the axiom's stated content, is unitary evolution; and no-signaling holds in the restricted sense of preserved commuting-observable statistics for a fixed record partition. Graded as the companion grades it: given the kernel, the observables axiom is a theorem of the postulates, replaced by a soundness theorem, and the projection postulate is the representation of a stipulated record-compatible restriction, which the validity postulate alone does not select; the state space, the superposition principle, composite systems, and unitary evolution are the kernel axiom's own content, read off the assignment; and the Born rule is absorbed under the axiom's calibration clause, the amplitudes being recovered from cluster weights by polarization, at each configuration by the companion's lemma and, up to the basis-phase gauge of the configurations and one common complex conjugation per recombination-connected component, across configurations by its stated global claim, and never held as objects. That is the sense in which the *formal* measurement problem is addressed as a theorem of the set given the kernel (the axiomatization companion's recovery theorem [9], and the long paper's statement against Maudlin's trilemma [4]).

*What remains outside the result.* Closure under arbitrary complex combinations is left open; completeness of the observable representation, every operator of interest arising from a record construction, is not claimed; the completeness of the state and observable representations, the continuum extension, and spin lie outside the theorem's grade; and the physical problem is relocated to the realization bridge, as §10 says. Its own derivation is assigned, explicitly, to a successor program—on the long paper's analogy, the Nexus occupies the position Newton's law of gravitation held beside his three laws: empirically load-bearing, conceptually awaiting its deeper motivation [4].

What the kernel already buys, checkably, is more than an interpretation usually offers. The per-polarizer $\cos^2/\sin^2$ weights are not stipulated apparatus-by-apparatus: they arrive as the kernel's own two-node overlaps once outcomes are fixed as determinate nodes, and full polarizer chains—interposed-polarizer revival included—fall out as sequential pruning residue, the Malus law arriving as the axiom's consequence rather than an input. The per-outcome decomposition and polygonal geometric-phase structure that the standard sequential-measurement treatment historically inserted by hand is likewise a theorem given the kernel (the Bloch sphere as software on Aethic hardware, per the signature above). And the duality budget splits along the axiom line, the chord needing no kernel and the circle using it, as section 9.1 runs it in order. Bayesian conditioning, meanwhile, needed no kernel—which is the unification in one sentence: one reduced-form machine, two weightings, classical statistics under the trivial one and quantum statistics under the kernel.

The axiom's deepest consequence corrects a misconception: amplitudes are not primitive contents of any state. The weight law stands as the root, and the amplitude is its polarization, recovered from the weights at each configuration by the local polarization lemma and across configurations only as far as a stated claim reaches; the kernel's classical side is two-limited, the exclusive regime and the agreeing regime at unit visibility, with the interacting-but-patternless class between them.

And the axiom's classical side is two-limited, not one—a taxonomy worth drawing before it is claimed for anything. Classicality onsets through two independent mechanisms: *Aethic decoherence* (records render the cross-scenario element *invalid*, so the cross-terms die because their elements die), and *phase washing* (the action *difference* across a readout cell grows huge—a common offset to $S/\hbar$ changes nothing—the fringes become finer than any readout cell, and a cell that integrates over many of them shows none—the companion's readout-cell point: cells are exclusive unions of endpoints whose weights add—so the stationary-phase limit [13, 19] emerges with no visible fringes even where coherence

survives). Crossing the two yields four regimes:

| | **Phase visible** ($S/\hbar \sim 1$) | **Phase washed** ($S/\hbar \gg 1$) |
|---|---|---|
| **Coherent** (cross-element valid) | **A:** standard interference | **C:** macroscopic Aethic coherence—the taxonomy's fourth cell |
| **Decohered** (cross-element invalid) | **B:** decoherence | **D:** the classical world |

Three cells are the recognized ones; Class C—scenarios still interacting, no wave pattern—is the fourth, and it is not an addition to the axiom but its direct implication: the Nexus weighting, applied where records are absent and the action is large, simply names the cell. The four-regime crossing is offered here as taxonomy; the companion develops the regime, its guards, and its open trigger construction in full [4].

### 9.1 The Two-Tag Deduction in Brief

Two numbers are computed for one experiment, visibility $V$ and distinguishability $D$, and the quantum paper carries the derivation [9]; the brief states the apparatus, the principal relations and the result.

The chord first, because it needs no kernel. A classically random tag is a disagreeing apparatus superposition: a convex split of the ensemble into a marked part at weight $p$—the record has stated the exclusion, the paths compose as separate weighted alternatives, $D = 1$, $V = 0$—and an unmarked part at weight $1 - p$—no record, one agreeing pair, $D = 0$, $V = 1$. Weights add: $D = p$, $V = 1 - p$, and $V + D = 1$. The straight line is not an additional complementarity postulate; it is the geometry of mixing an interference-bearing agreeing sector with a record-bearing disagreeing sector, and it uses the framework's convexity and nothing from the kernel. Coherent tagging is intermediate in a different way: every run carries both paths, and what varies is how alike their marker states are. Tag the slits with polarizations at $\pm\alpha$ from the incident axis, with marker overlap $c = \cos 2\alpha$—how much of the two markers is the same, and so how much path coherence remains available. Partial overlap lies off the chord because the alternatives were never split into record-separated components. For pure markers the *unconditional* visibility is $V = c$ and at the optimal placement $D$ attains its supremum $D^\star = \sqrt{1 - c^2}$, so $V^2 + (D^\star)^2 = 1$: the visibility is the marker overlap, the kernel's own cross-term; the distinguishability is what is left of unity once that overlap is squared away, the standard optimal-discrimination result, imported here pending the companion's in-formalism supremum (section 9). An illustrative check at $\alpha = 30°$: $c = \frac{1}{2}$, $V = \frac{1}{2}$, $D = \frac{\sqrt{3}}{2}$, $\frac{1}{4} + \frac{3}{4} = 1$.

The concrete experiment now verifies that the distinction produces this geometry. Place one analyzer at angle $\theta$ before the screen. The construction is the framework's indexing at work: to each slit, a determinate node on the Bloch sphere (its necessary polarization state for transfer); the elementary objects are augmented paths, a spacetime route carrying a Bloch route; and the pair exists by or-pruning, all paths not passing through one node *or* the other removed. Every trigonometric factor below is the Bloch face's contribution, and every $x$-dependence is spacetime's. While the which-tag attribute stays blank in principle, the two tagged routes stand as a single agreeing superposition; the analyzer's pass-or-block outcome is a record—here the disagreeing regime enters—and reality splits there into the two ports as a disagreeing superposition whose Aethic weights are the readout probabilities, $p_\theta = \frac{1}{2}\big(1 + c\cos(2\theta)\big)$—three quarters and one quarter at $\alpha = 30°$ with the analyzer vertical. Within each port the agreeing pair survives—passing did not resolve the which-tag attribute, since both tags project onto the port with nonzero amplitude—so the pair stands tilted, and amplitudes compose within the port: writing $c_1 = \cos(\theta - \alpha)$ and $c_2 = \cos(\theta + \alpha)$,

$$I_\theta(x) \;\propto\; c_1^2 + c_2^2 + 2\,c_1 c_2 \cos\big(k\,\Delta(x)\big), \qquad V_\theta = \frac{2\,|c_1 c_2|}{c_1^2 + c_2^2},$$

full fringes at the erasing angle, contrast falling exactly to $c$ at the Bloch-orthogonal straddle (the sphere doubling physical angles, so forty-five degrees on the bench), anti-fringes at the complement—the contrast a pure Bloch-face ratio, the stripe geometry pure spacetime.

The analyzer does not create distinguishability; it chooses the direction on the sphere along which the markers' existing difference is read out. The setting that extracts the most is *not* the tag basis: the ensemble capacity $D = |\sin 2\theta| \sin 2\alpha$—the gap between the two tags' answers $c_1^2$ and $c_2^2$ to the analyzer's question—peaks at $\theta = 45°$, the straddle just named, where the tags' projections *on the sphere* onto the analyzer axis are equal and opposite, and reaches $\sin 2\alpha = \sqrt{1 - c^2}$, the trace distance of two states of overlap $c$, Helstrom's success probability being $\frac{1}{2}(1 + D^\star)$; this is the standard result, imported as such, and it is what makes $D^\star$ a genuine supremum over placements rather than a reading at the obvious node (the tag basis itself, $\theta = \alpha$, gives only $\sin^2 2\alpha$; the port-normalized predictability $P_\theta$—Greenberger and Yasin's $P$ [24], a predictability rather than a distinguishability, since after the analyzer no marker remains—reaches 1 trivially at $\theta = 90° - \alpha$, where one tag is simply blocked, and is not the supremum quantity). The erasing angle, $\theta = 0$, where the tags project equally with the same sign and the fringes are full, is that setting's conjugate axis: extraction at forty-five degrees on the bench, erasure at zero, one operation under two cuts. The two port quantities are then two components of one normalization identity: with $P_\theta = |c_1^2 - c_2^2|/(c_1^2 + c_2^2)$,

$$(2c_1c_2)^2 + (c_1^2 - c_2^2)^2 \;=\; (c_1^2 + c_2^2)^2 \qquad \Longrightarrow \qquad V_\theta^2 + P_\theta^2 \;=\; 1,$$

the interference term standing as $V$ and the record-separation term as $D$—a second exact circle, port-normalized, distinct from the roadmap's unconditional one. The quadratic and the convex are then the kernel face and the weight face of a single setup: at weight grade the record sector splits linearly, $c_D = D$ and $c_A = 1 - D$ (the total-variation decomposition of the optimal marker measurement's statistics), while the square map joins the faces by relating the kernel's overlap to the weight face's capacity, $c^2 + D^2 = 1$—two relations, not one relation read twice. Across the ports, weights add: the two cross-terms sum to $2\cos(2\alpha)$ by the cosine subtraction formula, so the unconditional pattern has visibility $c$ at every analyzer angle—the analyzer redistributes record information between the ports without changing the unconditional pattern, no-signaling arriving as an identity rather than a check (placement column, per section 6). The linear split is a fact about the record statistics—the coupling-lemma decomposition of the optimal marker measurement's two outcome distributions into a common part of weight $1 - D$ and disjoint parts of weight $D$—and not a decomposition of the path cluster into an exclusive element and an agreeing pair: such a split would place the unconditional visibility on the chord at $1 - D$, whereas it stands at $c = \sqrt{1 - (D^\star)^2}$ at every analyzer angle, the loss at the erasing angle being the block port's anti-fringes canceling the pass port's fringes, both ports agreeing pairs. Three validities hold jointly for every survivor: the system Aethus of the setup, the photon's own case, and the Nexus's agreeing-versus-disagreeing structure. And the un-guessable remainder is no abstraction: the agreeing component's screen face is the erased-port conditional pattern, identically in the analyzer angle—one residual object under two cuts, the framework's surviving pair and the laboratory's sorted statistics the same thing. The full deduction, assumptions inventoried, is the companion's [4].

By status: the chord $1 - D \le V$ is framework-native and kernel-free, the circle $V^2 + D^2 = 1$ is the kernel's, and the two-tag configuration is where the qualitative classification and the quantitative prediction meet in one apparatus.

## 10 The Price, and the Close

The ledger is Table 1: each claim with its assumptions, owner paper and status, and nothing in this overview reads above its grade there.

The central outstanding physical construction, alongside the mathematical obligations Table 1 lists, is whether an observer-grade record structure can ever be kept from exporting its records, the premise under which the observer applications and the coherent-device exclusion hold, which is where the framework's divergence from universal unitarity lies and what a physical realization must settle by specifying the candidate configurations, their proper descendants and the record incompatibilities on which the third postulate operates.

The close is the trip run backwards in one paragraph. A single question generated everything: *what must be said about the universe so that quantum physics is not out of place?* Part one answered it for modality—alternatives brought home, the ordering deleted, permanence forced—and the union

move answered it for logic—exclusivity manufactured, interaction the default—and the two regimes the question demanded fell out as the two superposition kinds, with the powerset theorem drawing the boundary between them wherever the information sits. Modality itself ends the trip upgraded: counterfactuals are, on this accounting, not ways of speaking but structural features of the block—weighted, pruned, and measured.[10] Underneath both answers is the one move this paper promised to run and has now run twice: *determinacy is a two-place relation between an attribute and an information state, never a one-place property of spacetime-located facts.* The long paper [4] develops everything here at full formality—the semantics with its complete mode structure, the postulates with their fixed-point characterization, the Nexus weighting and its open kernel program, and the wider consequences from causation to cosmology—and the reader who wants the argument at its native resolution is warmly directed there. The short version ends where it began: one move, and a universe in which quantum physics is not out of place.

**Acknowledgments.** The framework, its extrusion principle (August 2022), and all fundamentals are the author's alone—including the one proto-form that predates the framework proper: the Attribute Set of 2021, whose no-privileged-decomposition property the companion names Decomposition Neutrality and dates in its genealogy (the journal entry that coined the term, December 7, 2021, is reproduced in the companion's genealogy—an attested date ahead of the deposited chain below, carried by the companion rather than staked here). To be specific, because the point admits no ambiguity: the three postulates, and their applications to time, to logic, to probability, and to the double-slit cases, were established by the author by May 2023, the derivations outlined end-to-end in the author's own terminology—a formalism invented from inductive reasoning on the nature of time together with the double-slit phenomenology, before encountering Wigner's friend, relational quantum mechanics, or any modal semantics. This is checkable rather than merely attested, and the primary documents are public.[11] The 2023 manuscripts stand in their deposited files—the January 28, 2023 intermediate print and the May 20, 2023 priority instrument—both restored verbatim, phrase-marked to the formal structures they anticipate. The complete formalism was deposited at PhilArchive in November 2024, nineteen months before any AI involvement, with the successive deposits (July 2025, January 2026, April 2026), frozen and dated by the archive rather than the author, dating the later-coined terms—the reduced form, the Quantum Nexus, the Powerset Invalidation Theorem—to the author, ahead of any exchange. A platform-dated spoken exposition of October 19, 2023 delivers the third postulate in its finished two-quantifier form thirteen months before the preprint. The continuity is laid out, print by print, in the companion paper [4]. The AI assistance of June 2026 was recognition and translation, not construction: it identified the standing fields the author's structures turned out to inhabit (most notably the counterfactual semantics of this paper's title), and converted the already-outlined derivations into their standard terminology; the formulation of the modal semantics was translated from the author's principle and worked instances. This compression was drafted and revised under multi-model adversarial review; where review supplied formal devices rather than translations—the scope menu, the pocket-wise operator—they remain credited in the companion paper's provenance notes. Every structure on the Aethic side of every correspondence asserted in this paper antedates the assistance in the public record.

© 2026. Licensed under CC BY-NC-SA 4.0.

# References

[1] Ernest W. Adams. *The Logic of Conditionals: An Application of Probability to Deductive Logic.* Dordrecht: D. Reidel, 1975.

[10] A boundary that matters for the belief-revision reader: what is structural is the *residue*; the conditional verdict itself is evaluated *at* books and never stored *in* them as an entry. Verdicts are not bookable attributes, which is the block that keeps the Ramsey-test triviality results [21] from getting started.

[11] The chain, linked in order: the January 2023 print, `https://aethus.org/aethic-code/files/Jan2023Aethic.pdf`; the May 2023 print, `https://aethus.org/aethic-code/files/Original_Aethic.pdf`; both restored verbatim and phrase-marked, `https://aethus.org/aethic-code/history/marked_original.html`; the PhilArchive version trail, `https://philarchive.org/versions/BENARA-5`; and the October 2023 spoken exposition, `https://youtube.com/playlist?list=PL1NA36JDv_T_1gJnzWhevIafOe0x4Q1yC`.

[2] Carlos E. Alchourrón, Peter Gärdenfors, and David Makinson. "On the Logic of Theory Change: Partial Meet Contraction and Revision Functions". In: *Journal of Symbolic Logic* 50.2 (1985), pp. 510–530.

[3] Markus Arndt et al. "Wave–particle duality of $C_{60}$ molecules". In: *Nature* 401 (1999), pp. 680–682.

[4] Ajax Benander. "Aethic Reasoning: A Comprehensive Solution to the Quantum Measurement Problem". In: (Nov. 2024). Alternative URL: https://philsci-archive.pitt.edu/id/eprint/25713. URL: https://philpapers.org/rec/BENARA-5.

[5] Ajax Benander. *Causality from Indexed Information States: Screening, Intervention, and the Bell Obstruction*. Preprint. 2026. URL: https://philpapers.org/rec/BENCWT-4.

[6] Ajax Benander. *Counterfactual Evaluation by Perturbation of Information States: Deleting the World-Similarity Ordering, Not Refining It*. Preprint. 2026. URL: https://philpapers.org/rec/BENCEB-2.

[7] Ajax Benander. *Determinacy as a Two-Place Relation: The Category Mistake in Quantum Interpretation, and the Dissolution of the Paradox Set*. Preprint. 2026. URL: https://philpapers.org/rec/BENTRO-72.

[8] Ajax Benander. *Newcomb's Problem Without a Fixed Past: A Decision Rule From Observer-Indexed Determinacy*. Preprint. 2026. URL: https://philpapers.org/rec/BENNPW.

[9] Ajax Benander. *Quantum Mechanics from Records: Measurement Structure on a Finite Path-Kernel Domain*. Preprint. 2026. URL: https://philpapers.org/rec/BENTQB.

[10] Ajax Benander. *The Weight Algebra of Aethae: A Two-Layer Commutative Semiring for Superposition and Pruning*. Preprint. 2026. URL: https://philpapers.org/rec/BENTWA-5.

[11] Garrett Birkhoff and John von Neumann. "The logic of quantum mechanics". In: *Annals of Mathematics* 37.4 (1936), pp. 823–843.

[12] Ivano Ciardelli, Linmin Zhang, and Lucas Champollion. "Two switches in the theory of counterfactuals: A study of truth conditionality and minimal change". In: *Linguistics and Philosophy* 41.6 (2018), pp. 577–621.

[13] P. A. M. Dirac. "The Lagrangian in quantum mechanics". In: *Physikalische Zeitschrift der Sowjetunion* 3 (1933), pp. 64–72.

[14] Stephan Dürr, Thomas Nonn, and Gerhard Rempe. "Origin of Quantum-Mechanical Complementarity Probed by a 'Which-Way' Experiment in an Atom Interferometer". In: *Nature* 395 (1998), pp. 33–37. DOI: 10.1038/25653.

[15] Albert Einstein, Boris Podolsky, and Nathan Rosen. "Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?" In: *Physical Review* 47.10 (1935), pp. 777–780.

[16] Avshalom C Elitzur and Lev Vaidman. "Quantum mechanical interaction-free measurements". In: *Foundations of Physics* 23.7 (1993), pp. 987–997.

[17] Berthold-Georg Englert. "Fringe Visibility and Which-Way Information: An Inequality". In: *Physical Review Letters* 77.11 (1996), pp. 2154–2157. DOI: 10.1103/PhysRevLett.77.2154.

[18] Yaakov Y. Fein et al. "Quantum superposition of molecules beyond 25 kDa". In: *Nature Physics* 15 (2019), pp. 1242–1245. DOI: 10.1038/s41567-019-0663-9.

[19] R. P. Feynman. "Space-Time Approach to Non-Relativistic Quantum Mechanics". In: *Reviews of Modern Physics* 20 (1948), pp. 367–387.

[20] Daniela Frauchiger and Renato Renner. "Quantum Theory Cannot Consistently Describe the Use of Itself". In: *Nature Communications* 9 (2018), p. 3711.

[21] Peter Gärdenfors. "Belief Revisions and the Ramsey Test for Conditionals". In: *The Philosophical Review* 95.1 (1986), pp. 81–93.

[22] Peter Gärdenfors. *Knowledge in Flux: Modeling the Dynamics of Epistemic States*. Cambridge, MA: MIT Press, 1988.

[23] Daniel M. Greenberger and Allaine Yasin. "Simultaneous Wave and Particle Knowledge in a Neutron Interferometer". In: *Physics Letters A* 128.8 (1988), pp. 391–394. DOI: 10.1016/0375-9601(88)90114-4.

[24] Daniel M. Greenberger and Allaine Yasin. "Simultaneous wave and particle knowledge in a neutron interferometer". In: *Physics Letters A* 128.8 (1988), pp. 391–394.
[25] Adam Grove. "Two Modellings for Theory Change". In: *Journal of Philosophical Logic* 17.2 (1988), pp. 157–170.
[26] Radu Ionicioiu and Daniel R. Terno. "Proposal for a quantum delayed-choice experiment". In: *Physical Review Letters* 107.23 (2011), p. 230406.
[27] Simon Kochen and Ernst P. Specker. "The problem of hidden variables in quantum mechanics". In: *Journal of Mathematics and Mechanics* 17.1 (1967), pp. 59–87.
[28] Angelika Kratzer. *Modals and Conditionals: New and Revised Perspectives*. Oxford Studies in Theoretical Linguistics. Revised and extended treatment; useful if citing the more recent formulation. Oxford: Oxford University Press, 2012.
[29] David Lewis. *Counterfactuals*. Oxford: Blackwell, 1973.
[30] David Lewis. "Ordering Semantics and Premise Semantics for Counterfactuals". In: *Journal of Philosophical Logic* 10.2 (1981), pp. 217–234.
[31] Asher Peres. "Unperformed experiments have no results". In: *American Journal of Physics* 46.7 (1978), pp. 745–747. DOI: 10.1119/1.11393.
[32] Matthew F. Pusey, Jonathan Barrett, and Terry Rudolph. "On the reality of the quantum state". In: *Nature Physics* 8 (2012), pp. 475–478.
[33] Hilary Putnam. "Is Logic Empirical?" In: *Boston Studies in the Philosophy of Science, Volume 5*. Ed. by Robert S. Cohen and Marx W. Wartofsky. Reprinted as "The Logic of Quantum Mechanics" in *Mathematics, Matter and Method* (1975). Dordrecht: D. Reidel, 1968, pp. 216–241. DOI: 10.1007/978-94-010-3381-7_5.
[34] Marlan O. Scully, Berthold-Georg Englert, and Herbert Walther. "Quantum optical tests of complementarity". In: *Nature* 351 (1991), pp. 111–116.
[35] Robert W. Spekkens. "Evidence for the Epistemic View of Quantum States: A Toy Theory". In: *Physical Review A* 75 (2007), p. 032110.
[36] Robert W. Spekkens. "Negativity and Contextuality are Equivalent Notions of Nonclassicality". In: *Physical Review Letters* 101 (2008), p. 020401. DOI: 10.1103/PhysRevLett.101.020401.
[37] Robert C. Stalnaker. "A Theory of Conditionals". In: *Studies in Logical Theory*. Ed. by Nicholas Rescher. Oxford: Blackwell, 1968, pp. 98–112.
[38] John Archibald Wheeler. "Law Without Law". In: *Quantum Theory and Measurement*. Ed. by John Archibald Wheeler and Wojciech Hubert Zurek. Princeton: Princeton University Press, 1983, pp. 182–213.
[39] Eugene P. Wigner. "Remarks on the mind-body question". In: *The Scientist Speculates*. Ed. by Irving John Good. Heinemann, 1961, pp. 284–302.
[40] Wojciech H. Zurek. "Quantum Darwinism". In: *Nature Physics* 5 (2009), pp. 181–188.